

Network Multipliers in Gravity Models

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Abstract

This paper proposes a new gravity specification grounded in countries' network-leveraging features (e.g., logistics hubs). Building on spatial autoregressive models, we endogenize trade costs as a function of a network of country proximities. The resulting specification induces interdependence among trade flows and pair-specific heterogeneity, and nests the conventional iceberg cost as a special case when countries do not leverage network connections. For estimation, we use PPML and develop robust inference accommodating heteroskedasticity and unknown error correlations. We also propose a fast algorithm for the large-dimensional network-multiplier transformation. Empirically, we find evidence of endogenous trade costs and substantially improved fit relative to the conventional gravity equation (up to 30% in-sample fit), implying pair-specific heterogeneity missed by existing models. Counterfactual analysis shows that a bilateral increase in trade costs generates sizable reallocations of import shares across many third countries once network interactions are accounted for.

Keywords: Origin-destination flows, Gravity equation, Network, Iceberg cost, Spatial autoregressive model, Poisson pseudo-maximum likelihood estimation, Eigenvalues
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1 Introduction

How do bilateral trade policy shocks—such as sanctions or supply-chain disruptions—propagate through the trade network to affect third-country trade patterns? Recent episodes such as

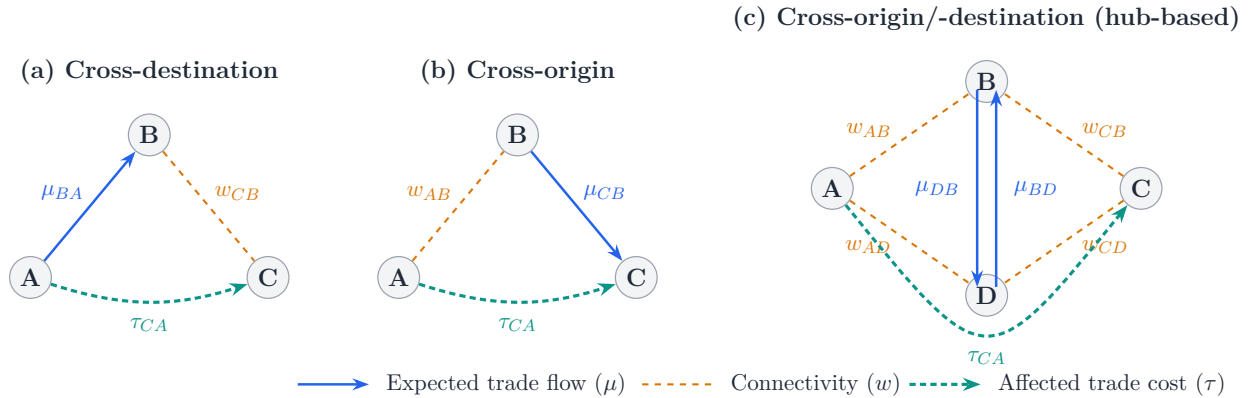
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the China–US trade war suggest that impacts extend beyond nearby partners (e.g., Korea and Japan) to more remote countries. Trade policy shocks are rarely confined to the targeted pair: in an interconnected network, changes in bilateral trade costs propagate through origin-, destination-, and hub-based linkages, reshaping trade patterns far beyond the affected dyad. This paper asks whether gravity-based policy models systematically understate the scope and consequences of trade interventions. We show that when trade costs respond endogenously to network-wide reallocations, bilateral shocks generate large third-country reallocation and concentration effects that standard gravity largely misses.

Gravity equations are the workhorse for explaining bilateral trade flows. In the structural gravity framework, specifying trade costs is crucial because it shapes both partner composition and traded quantities. The iceberg-cost assumption—under which a fraction of the shipped good “melts” in transit—is the canonical choice: it is microfounded in standard models, enters trade shares multiplicatively, and yields clean identification and counterfactual mappings. Empirically, distance and tariffs proxy for this ad valorem cost. The conventional formulation treats trade costs as exogenous and unavoidable, and assumes they are formed independently across country pairs even when pairs are geographically or logistically connected. This is distinct from multilateral resistance, which captures general-equilibrium expenditure reallocation; our focus is instead on trade costs becoming network-dependent across origins, destinations, and third-country hubs.

Figure 1: Network linkages in the formation of bilateral trade costs



This paper revisits the conventional view by asking whether countries respond to bilateral trade costs by leveraging information about expected third-party flows—for example, by routing or coordinating shipments. Higher expected flows may signal reliable routes and scale economies, but can also raise costs when capacity constraints bind. Fig. 1 summarizes the mechanisms. Consider countries A, B, and C. If $A \rightarrow B$ is expected to be large and B is

strongly linked to C (panel a), routing via B can share schedules and fixed logistics, lowering the effective A→C cost. If instead A and B are closely connected and B→C is expected to be large (panel b), A can bundle A→B shipments with B→C services, again reducing the effective cost to C. More generally, expected flows among third parties (e.g., B↔D), together with strong proximities (A–D and B–C), can make multi-leg routes such as A→D→B→C or A→B→D→C cost-effective (panel c). Conversely, when shipping capacity is constrained, large expected flows on adjacent routes (e.g., A→B, B→C, B↔D) may increase the effective cost of A→C. Thus, trade costs can depend on expectations about third-party flows, generating cross-origin and cross-destination linkages that a purely pairwise iceberg view misses. These considerations motivate a framework that endogenizes trade costs through network-based linkages.¹

We develop a microfounded gravity model in which trade costs are endogenized using a spatial autoregressive (SAR) operator structure to capture network interactions. The core hypothesis is that expected trade flows, filtered through countries’ connectivity, shape bilateral costs, rather than costs being driven solely by geographic distance. Let y_{ij} denote the observed flow from j to i and μ_{ij} its systematic component. The model has three stages. Stage 1 constructs a long-run connectivity matrix $W = (w_{ij})$, where w_{ij} measures how strongly i is linked to j in trade-relevant proximity, capturing the importance of nearby routes and hubs for each pair’s shipping options. Stage 2 specifies ad-valorem trade costs as the product of a network-driven component (a function of expected flows and W) and standard bilateral covariates. Stage 3 characterizes equilibrium trade flows via a CES demand system à la Anderson and van Wincoop (2003).

The endogenous component of trade costs is multiplicative (ad-valorem) and depends on geometric averages of expected flows weighted by connectivity $\{w_{ij}\}$. Three terms capture distinct network channels: (i) outflows from a common origin (A→ ·) generate cross-destination linkages (routing/backhaul); (ii) inflows to a common destination (· →C) generate cross-origin linkages (coordination toward the same market); and (iii) third-party flows capture hub-and-spoke and multi-leg routing effects.

We map the theoretical equilibrium to the data and estimate a gravity equation in which bilateral trade costs depend on (i) network-spillover elasticities—capturing how connected flows across origins, destinations, and third-country hubs shape effective trading costs—and (ii) standard elasticities on bilateral shifters such as distance and borders. Network spillovers

¹Hub ports (e.g., Singapore or Dubai) illustrate how third-party flows generate network economies by pooling shipments across origins and destinations. In container shipping, eastbound flows (Asia→US) typically exceed westbound flows, creating costly empty-container repositioning; expansions in westbound exports create backhaul opportunities that lower unit costs on the complementary leg. Similar backhaul and coordination mechanisms arise in air cargo and long-haul trucking.

can reduce costs through coordination, routing, and bundling, but can also raise them when capacity constraints bind; thus, their signs distinguish efficiency amplification from congestion transmission. Crucially, these spillovers generate systematic pair-specific heterogeneity that conventional gravity would leave in the residual. Empirically, the model uncovers pair-specific variation correlated with proximity to common routes and hubs. Setting network spillover elasticities to zero nests the standard iceberg-cost gravity model.

To ensure a well-defined econometric model, we characterize equilibrium trade flows and their uniqueness. Equilibrium is pinned down by two contraction mappings: (i) conventional multilateral resistance terms and (ii) the network SAR operator. The resulting equilibrium admits an SAR structure in an exponential functional form. This is a semi-reduced form obtained by inverting only the network SAR operator, since the fixed-effect components are implicit functions of the systematic trade flows: they embed multilateral resistance, i.e., expenditure-share aggregations weighted by endogenized trade costs. The semi-reduced form highlights a simple propagation logic: a shock to one bilateral pair changes expected flows elsewhere, which feeds back into trade costs and reallocates trade across the network. The econometric model then represents y_{ij} as the expected flow μ_{ij} plus an error term.

We consider the Poisson pseudo-maximum likelihood (PPML) estimator. This method only requires a correctly specified conditional mean of y_{ij} , so we allow for arbitrary heteroskedasticity and correlation in the error terms and construct heteroskedasticity- and spatial-correlation-robust (HAC) standard errors. Robust inference is crucial because network dependence in the conditional mean may also induce spatial dependence in the errors, so we remain agnostic about the correlation structure. We state identification conditions and derive the PPML estimators' asymptotic distributions. Relative to conventional gravity, network feedback further restricts the admissible parameter space. Monte Carlo simulations show small bias and accurate coverage in realistic sample sizes.

Methodologically, we develop a spectral algorithm to compute transformations by the network multiplier matrix (the inverse of the network SAR operator), which is all that is required for estimation. With n countries, this multiplier is $N \times N$ with $N = n^2$, so direct inversion is infeasible. Our algorithm exploits the eigendecomposition of W to simultaneously diagonalize the three Kronecker-type network operators capturing cross-destination, cross-origin, and joint origin-destination linkages. Transformations by the multiplier then reduce to simple elementwise operations in the eigenbasis, dramatically lowering computational cost. In our application ($n \simeq 150$), the spectral algorithm is over four orders of magnitude faster than direct inversion, making estimation feasible for moderately large n .

Using our model, we study how network effects in bilateral trade flows shape effective

trade costs. We focus on two benchmark periods that capture the maturation of the global trade network: Period 1 (1997, a developing network period) and Period 2 (2016, a mature network period). Because the econometric model is cross-sectional, we estimate it separately for each period. In both periods, all three spillover channels are positive and statistically significant, indicating that bilateral flows are mutually reinforcing within the trade network. The hub-related (third-country) component is especially pronounced, pointing to a central role for shared hubs in amplifying trade linkages. Relative to a conventional iceberg-cost gravity model, incorporating these spillovers yields substantial likelihood-based fit gains (up to about 28%), highlighting the first-order importance of third-party proximities and network-mediated trade costs.

We show that allowing bilateral trade costs to respond endogenously to network-wide reallocations materially changes the effects of trade policy. In conventional structural gravity, a bilateral cost shock can affect third-country trade shares through multilateral resistance, but trade costs remain pair-specific and do not inherit network linkages. In our framework, changes in flows along connected origins, destinations, and third-country hubs feed back into effective bilateral costs, adding a propagation mechanism beyond multilateral resistance.

In counterfactuals motivated by the U.S.–China trade war, these network interactions imply that a bilateral increase in trade costs triggers sizable reallocations of import shares across a broad set of third countries, reshaping import concentration and diversification even in economies not directly targeted. This structural distinction explains why our model predicts larger and more widely dispersed third-country adjustments than conventional gravity benchmarks.

These results underscore that the global trading system is better viewed as an interconnected network than as a set of independent bilateral relationships. By allowing bilateral trade costs to be network-dependent, our framework links policy shocks to both direct bilateral responses and broader third-country reallocations operating through connected origins, destinations, and hubs. The policy implication is straightforward: trade interventions should be evaluated in a network-aware manner—anticipating rerouting and spillovers—and their incidence assessed not only at the targeted pair but across the wider set of connected economies. Analyses based solely on bilateral wedges and multilateral resistance can understate both the magnitude and the incidence of trade shocks, especially for economies tied to major hubs.

1.1 Our Contribution and Related Literature

Our contribution is to establish a spatial econometric framework for gravity models with network interactions operating through endogenized trade costs. We draw on spatial econometrics and trade/spatial economics and link them via endogenous, network-dependent trade costs.

First, we contribute to the endogenization of trade costs by introducing countries’ network-leveraging features in gravity models. This goes beyond the canonical iceberg-cost formulation (Samuelson Samuelson, 1952, 1954), which treats bilateral trade costs as exogenous and pair-specific. Since early gravity models (Tinbergen, 1962; Anderson, 1979; Helpman and Krugman, 1985), the iceberg framework has underpinned modern structural gravity, including (i) Armington-based models (McCallum, 1995; Anderson and van Wincoop, 2003; Arkolakis et al., 2012; Allen and Arkolakis, 2022; Wong, 2022), (ii) Ricardian models (Eaton and Kortum, 2002; Caliendo and Parro, 2015; Lind and Ramondo, 2023), and (iii) heterogeneous-firm models with fixed costs and extensive margins (Melitz, 2003; Chaney, 2008; Helpman et al., 2008; Morales et al., 2019). While third-country spillovers are central even in conventional gravity, they operate through multilateral resistance and are absorbed by country fixed effects in econometric implementations, limiting what the model can say about heterogeneity arising from relationships themselves.

A related strand of the literature studies *interdependence* in trade and transport networks, including Brancaccio et al. (2020) and Fuchs and Wong (2024), which embed network dependence in microfoundations of the transportation sector. In their settings, routing, capacity constraints, and decentralized shipping decisions jointly determine transport costs and the dependence across trade links or transport routes. The network topology (e.g., geography and infrastructure) is treated as a primitive, while equilibrium traffic patterns and congestion generate interdependence in costs. By contrast, our paper takes a given country connectivity matrix—constructed from long-run proximities and pre-determined linkages—as an econometric object and quantifies how this fixed network structure transmits and amplifies interdependence across bilateral trade flows.

To keep the focus on trade-cost formation, we retain the workhorse Armington-based structural gravity framework (Anderson and van Wincoop, 2003) to describe trade-flow determination. This demand-side foundation provides a transparent mapping from multiplicative trade-cost wedges to aggregate bilateral flows, making our network channel directly estimable with gravity data using PPML. In this sense, our departure from the canonical framework lies in how trade costs are formed and become interdependent across pairs, not in the demand system that allocates expenditure across origins. The resulting model augments

standard gravity with an additional set of parameters governing network-induced trade costs (beyond the usual bilateral shifters), while preserving the familiar general-equilibrium structure highlighted in universal gravity results (Allen et al., 2020). Identification comes from the restriction that each bilateral cost loads on origin-, destination-, and hub-based network aggregates constructed from the connectivity matrix, generating systematic cross-pair variation distinct from bilateral cost shifters.

Second, our framework builds on the spatial econometrics literature, which models interdependence across space or networks through spatial-lag structures. Our specification is motivated by SAR models (Cliff and Ord, 1995; Ord, 1975; Lee, 2004, 2007; LeSage and Pace, 2008), in which outcomes depend on neighboring outcomes. Recent extensions to origin–destination (OD) flows (Jeong et al., 2023; Jeong and Lee, 2024) and international trade applications (Behrens et al., 2012; Jin et al., 2023) further advance the modeling of bilateral interdependence. As a computational contribution, we exploit the network structure to compute the linear transformations implied by the LeSage–Pace (2008) network multiplier efficiently.

Existing SAR models for OD flows, however, typically rely on log-linearized specifications and cannot directly accommodate zero trade flows. As noted by Santos Silva and Tenreyro (2006), log transformations in gravity can induce bias and require ad hoc treatments of zeros. Building on our microfounded structure, we formulate the model in levels and estimate it using PPML (Gourieroux et al., 1984; Santos Silva and Tenreyro, 2006; Head and Mayer, 2014; Santos Silva and Tenreyro, 2022), in line with recent developments (Nagengast and Yotov, 2025; Kwon et al., 2025). We derive an asymptotic distribution for the PPML estimator that is asymptotically bias-free, consistent with recent results for linear/nonlinear fixed-effects models (Kapetanios et al., 2021; Weidner and Zylkin, 2021; Fernandez-Val and Weidner, 2016, 2018; Chen et al., 2021). For inference, we adopt heteroskedasticity- and autocorrelation-consistent (HAC) methods for spatial data (Kelejian and Prucha, 2007; Kim and Sun, 2011; Conley et al., 2023).

Third, our empirical results speak directly to the recent trade-cost literature. Using gravity data and a connectivity-based structure, we find strong evidence that bilateral trade costs are network-dependent, consistent with transportation-sector microfoundations that endogenize trade costs through network interactions (Brancaccio et al., 2020; Wong, 2022). Our estimates also support heterogeneous dependence patterns across countries, rather than uniform linkages (Lind and Ramondo, 2023). Crucially, allowing for network dependence converts what conventional gravity models would treat as residual variation into systematic pair-specific heterogeneity in trade flows. Across the main phases analyzed, we find economically meaningful elasticities on linkages to nearby routes and hubs, consistent with recent

evidence on the role of hubs in trade and spatial reallocation (Ganapati et al., 2024). These findings underscore that trade policy evaluation and resilience analysis should be network-aware, paying attention to how shocks propagate through critical routes and hubs—a perspective aligned with broader insights from network-based transportation analyses (Allen and Arkolakis, 2022; Fuchs and Wong, 2024).

The rest of the paper is organized as follows. Sec. 2 presents the microfoundations and the resulting econometric specification, including estimation and inference. Sec. 3 develops the statistical analysis, reporting asymptotic properties and Monte Carlo evidence. Sec. 4 applies the framework to world trade flows from the Center for International Data at UC Davis. Sec. 5 concludes.

2 Model

We characterize a spatial model for an origin–destination (OD) flow y_{ij} , where j denotes the origin and i the destination. Suppose we observe $N = n^2$ flows, or $N = n(n - 1)$ when intra flows are excluded, collected in an $n \times n$ matrix Y or $\mathbf{y} = \text{vec}(Y)$.² We index observations by the OD unit $ij \equiv (i, j)$. We adopt the location setting of Jenish and Prucha (2009, 2012).

Assumption 2.1. Let $\mathcal{I} = \{1, \dots, n\}$ be the set of country indices. Each $i \in \mathcal{I}$ has a location $l(i) \in \mathcal{D}_n \subset \mathcal{D}$, where $\mathcal{D}$ is the set of all potential locations in $\mathbb{R}^d$. Assume $\lim_{n \rightarrow \infty} \#(\mathcal{D}_n) = \infty$ and $\min_{i \neq j} d(l(i), l(j)) \geq 1$, where $\#(\mathcal{D}_n)$ is the cardinality of $\mathcal{D}_n$, $l : \mathcal{I} \rightarrow \mathcal{D}$ is an injective location map, and $d(\cdot, \cdot)$ is a distance metric.

Assumption 2.1 is standard in the spatial econometrics literature. Beyond geographic space, $\mathcal{D}$ can be interpreted broadly as a characteristic space capturing countries’ economic or political proximity.

Traditional spatial autoregressive (SAR) models study observations (y_i, x_i) collected under Assumption 2.1, where y_i is an outcome and x_i a regressor vector. The SAR model specifies interdependence as

$$y_i = \lambda \sum_{j=1}^n w_{ij} y_j + x_i' \beta + v_i, \quad i = 1, \dots, n, \quad (2.1)$$

where $W = (w_{ij})$ is an $n \times n$ weight matrix, λ and β are parameters, and v_i is an error. If

²We use the *destination-centric ordering* (LeSage and Pace, 2008), under which y_{ij} is the (i, j) element of Y . This ordering is convenient for matrix notation and for viewing Y as a directed weighted network. It is also used in the trade literature, e.g., Eaton and Kortum (2002) and Head and Mayer (2014).

$S \equiv I_n - \lambda W$ is invertible, (2.1) has a unique solution. In addition, if $\rho_{\text{spec}}(\lambda W) < 1$,

$$y_i = \sum_{j=1}^n (S^{-1})_{ij} (x'_j \beta + v_j) = \sum_{j=1}^n (I_n + \lambda W + \lambda^2 W^2 + \cdots)_{ij} (x'_j \beta + v_j). \quad (2.2)$$

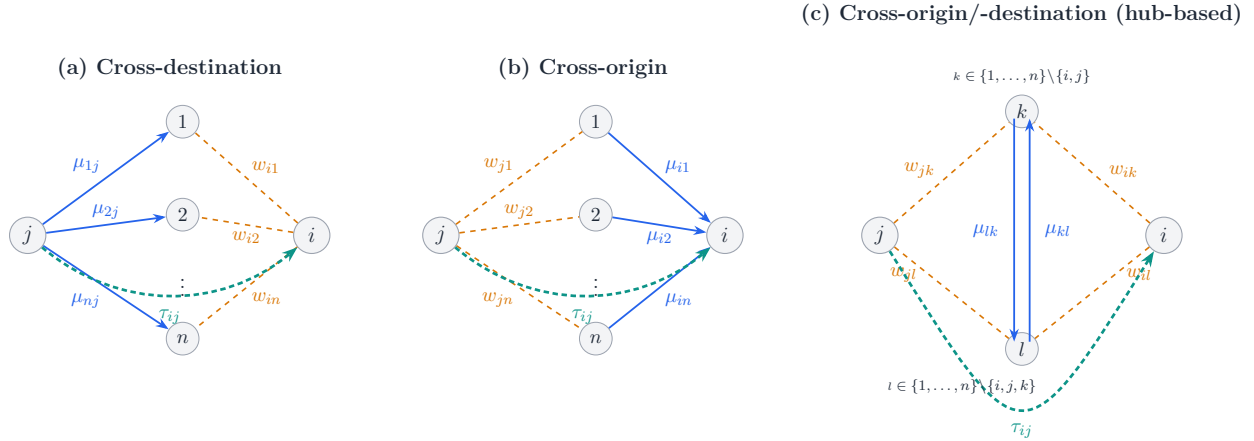
Thus, $\mathbb{E}(y_i \mid x) = \sum_j (S^{-1})_{ij} x'_j \beta$ and $\text{Var}(y_i \mid x) = \sigma^2 (S^{-1} S^{-1'})_{ii}$ when $\mathbb{E}(v_i \mid x) = 0$, $\text{Var}(v_i \mid x) = \sigma^2 > 0$ for all i , and $\text{Cov}(v_i, v_j \mid x) = 0$ for $i \neq j$. This representation highlights how interconnectivities generate spillovers and regional heterogeneity.

2.1 Theoretical foundation

This subsection provides microfoundations for our model. Our goal is to specify μ_{ij} —the systematic component/conditional mean of y_{ij} —by endogenizing trade costs.

We formalize the model in three stages. Motivated by transport-based theories in which trade costs respond to network conditions and traffic patterns (e.g., Brancaccio et al., 2020; Wong, 2022), we allow bilateral trade costs to depend not only on pairwise characteristics but also on network-linked neighborhood conditions, summarized by a connectivity matrix and expected flows. Fig. 2 illustrates the partner linkages underlying this connectivity structure.

Figure 2: Partner and hub neighborhoods for the target trade cost τ_{ij}



Stage 1 (long-run partner and hub selection). Consider a destination i and an origin j with $i \neq j$. Let w_{ik}^d and w_{jl}^o denote the stationary long-run frequencies with which i and j are paired with counterparties $k \in \mathcal{I} \setminus \{i\}$ and $l \in \mathcal{I} \setminus \{j\}$ in their respective trading neighborhoods. For a pairing opportunity ν ,

$$\Pr(k_{i\nu} = k) = w_{ik}^d, \quad \Pr(l_{j\nu} = l) = w_{jl}^o.$$

We collect these weights into row-stochastic matrices $W^d = (w_{ik}^d)$ and $W^o = (w_{jl}^o)$ with zero diagonals. They summarize relative long-run connectivity relevant for trade-cost formation rather than current bilateral trade flows. As illustrated in Fig. 2(a)–(c), these weights generate the three network neighborhoods—cross-destination, cross-origin, and hub-based—that enter the endogenous cost block in Stage 2. For the empirical application, we impose a single long-run proximity structure and set $W^d = W^o = W$. Thus, in our framework, W is a long-run network primitive, while endogeneity arises only in Stage 2 through expected-flow aggregation.

Remark 2.1 (Maritime search–matching). One economic rationale for this long-run connectivity kernel is provided by the spatial search–matching framework of Brancaccio et al. (2020) (BKP). In BKP, the optimizing agents are ships and exporters rather than countries: unmatched ships choose whether to stay or ballast elsewhere, and the resulting steady-state composition of the ship pool arriving at a destination induces a country-level connectivity structure. This gives a natural interpretation to our destination-side kernel W^d as a passive steady-state composition rather than an active country-level choice. Our use of a common two-sided kernel $W^d = W^o = W$ should therefore be viewed as a tractable reduced-form approximation, not a literal implication of BKP. Appendix 1.5 shows how a related country-level connectivity object can emerge from BKP’s ship-level equilibrium.

Assumption 2.2. Let $\tilde{W}$ be a nonnegative symmetric matrix with zero diagonal and positive row sums. Define

$$W = \text{Diag}^{\text{sum}}(\tilde{W})^{-1} \tilde{W}, \quad \text{Diag}^{\text{sum}}(\tilde{W}) = \text{diag} \left(\sum_{j=1}^n \tilde{w}_{1j}, \dots, \sum_{j=1}^n \tilde{w}_{nj} \right),$$

so that W is row-stochastic and may be asymmetric after normalization.

The common-kernel specification provides a parsimonious reduced-form summary of partner relevance. The framework can accommodate $W^d \neq W^o$, at the cost of additional primitives and computation.

Stage 2 (endogenous cost formation). Let τ_{ij} be a measure of a trade cost from j to i . We allow τ_{ij} to depend on network-linked trading activity around the origin and destination in addition to bilateral characteristics. We posit the following cost function that endogenizes trade costs.

Assumption 2.3. (i) For each ij , we assume

$$\tau_{ij}^+ = D_{ij,1}^{\tilde{\beta}_1} \cdots D_{ij,K}^{\tilde{\beta}_K},$$

where $D_{ij,k}$ ($k = 1, \dots, K$) represents a bilateral characteristic affecting τ_{ij} . $\tilde{\beta}_1, \dots, \tilde{\beta}_K$ are parameters. We assume that the baseline cost τ_{ij}^+ satisfies the triangle inequality: for arbitrary three countries i, j , and k , $\tau_{ij}^+ \leq \tau_{ik}^+ \cdot \tau_{kj}^+$.

(ii) If i chooses $k \in \{1, \dots, n\} \setminus \{i\}$ with probability w_{ik} and j selects $l \in \{1, \dots, n\} \setminus \{j\}$ with probability w_{jl} as partners (hubs), the trade cost from j to i through k and l is

$$\tilde{\tau}_{ij}(\boldsymbol{\mu}; k, l) = \mu_{kj}^{-\tilde{\lambda}_d} \mu_{il}^{-\tilde{\lambda}_o} \mu_{kl}^{-\tilde{\lambda}_w} \cdot \tau_{ij}^+, \quad (2.3)$$

where $\tilde{\lambda}_d$, $\tilde{\lambda}_o$, and $\tilde{\lambda}_w$ are coefficients, and $\boldsymbol{\mu} = (\mu_{11}, \mu_{21}, \dots, \mu_{n1}, \dots, \mu_{1n}, \mu_{2n}, \dots, \mu_{nn})'$. For analytical and computational convenience, we normalize $\mu_{ii} = 1$ for all $i = 1, \dots, n$.

(iii) i 's and j 's partner choices are independent, so the probability of using the route k and l is $w_{ik}w_{jl}$.

(iv) Then, the overall trade cost from j to i is defined as

$$\tau_{ij}(\boldsymbol{\mu}) = \exp(\mathbb{E}_W[\ln(\tilde{\tau}_{ij}(\boldsymbol{\mu}; k, l))]), \text{ where } \mathbb{E}_W(\cdot) = \sum_{k,l=1}^n w_{ik}w_{jl}(\cdot).$$

As a result, we have

$$\tau_{ij}(\boldsymbol{\mu}) = \tau_{ij}^e(\boldsymbol{\mu}) \cdot \tau_{ij}^+, \text{ where } \tau_{ij}^e(\boldsymbol{\mu}) = (\bar{\mu}_{.j}^i)^{-\tilde{\lambda}_d} (\bar{\mu}_i^j)^{-\tilde{\lambda}_o} (\bar{\mu}_{..}^{ij})^{-\tilde{\lambda}_w} \quad (2.4)$$

with $\bar{\mu}_{.j}^i = \prod_{k=1}^n \mu_{kj}^{w_{ik}}$; $\bar{\mu}_i^j = \prod_{l=1}^n \mu_{il}^{w_{jl}}$; and $\bar{\mu}_{..}^{ij} = \prod_{k,l=1}^n \mu_{kl}^{w_{ik}w_{jl}}$.

(v) All stages—**Stage 1**, **Stage 2**, and **Stage 3**—operate under no informational frictions, meaning that the expected flows in **Stage 2** are the same as the equilibrium flows in **Stage 3**, $\boldsymbol{\mu}^* = (\mu_{11}^*, \mu_{21}^*, \dots, \mu_{n1}^*, \dots, \mu_{1n}^*, \mu_{2n}^*, \dots, \mu_{nn}^*)'$.

Under these settings, **Stage 2** forms $\tau_{ij}(\boldsymbol{\mu}^*)$ for $i, j = 1, \dots, n$.

Assumption 2.2 illustrates properties of a country connectivity matrix (W) from **Stage 1**. The symmetry of the source matrix ($\tilde{W}$) is conventional and adopted for computational convenience (see Sec. 2.3). Assumption 2.3 describes how trade costs are endogenously shaped by information in trade networks. The three panels in Figure 2 correspond directly to the three network-based terms in equation (2.4). Unlike conventional models with exogenous τ_{ij} , here trade costs are themselves functions of expected flows weighted by countries' connectivities.

The component $\tau_{ij}^e(\boldsymbol{\mu})$ within $\tau_{ij}(\boldsymbol{\mu})$ captures the network-based part of trade costs:

1. Outflows from j ($\bar{\mu}_{.j}^i$) summarize common-origin, cross-destination linkages;

2. Inflows to i ($\bar{\mu}_{i.}^j$) summarize common-destination, cross-origin linkages; and
3. Hub-based flows ($\bar{\mu}_{..}^{ij}$) summarize joint cross-origin and cross-destination linkages.

Here, $\{w_{ik}\}_{k=1}^n$ in $\bar{\mu}_{.j}^i$ capture cross-destination linkages, $\{w_{jl}\}_{l=1}^n$ in $\bar{\mu}_{i.}^j$ capture cross-origin linkages, and both sets of weights appear in $\bar{\mu}_{..}^{ij}$ to capture their joint effect. A positive $\tilde{\lambda}_d$ implies network-based efficiency gains through cross-destination linkages, whereas a negative $\tilde{\lambda}_d$ reflects congestion effects when nearby routes or hubs operate at capacity. Analogously, $\tilde{\lambda}_o$ governs cross-origin linkages (e.g., coordination versus destination-side bottlenecks), and $\tilde{\lambda}_w$ captures hub-based routing effects (e.g., hub-and-spoke economies versus congestion on multi-leg paths).

Since our empirical focus is on inter-regional trade flows, we do not model intra-regional flows explicitly in the baseline specification. Accordingly, we impose the normalization $\mu_{ii} = 1$ for all i . Under this normalization, domestic cells remain multiplicatively neutral in (2.3): whenever $k = j$, $l = i$, or $k = l$, the corresponding domestic-flow term contributes a factor of one and therefore does not affect the route-specific trade cost. This device preserves the full n^2 notation and facilitates the efficient spectral computation in Sec. 2.3. An alternative implementation that excludes domestic cells from the network-cost aggregator via an elimination operator is discussed in Appendix 1.6.

Our specification nests the conventional iceberg cost specification since $\tau_{ij}(\boldsymbol{\mu}) = \tau_{ij}^+$ when $\tilde{\lambda}_d = \tilde{\lambda}_o = \tilde{\lambda}_w = 0$. In this case, cross-border arbitrage satisfies the triangle inequality, highlighting the effectiveness of geographic barriers (e.g., Eaton and Kortum (2002)). By contrast, in our general case, the triangle inequality need not hold, implying that countries can achieve effective cost reductions based on the expected trade network.

Stage 3 (trade flow formation). Given τ_{ij} from **Stage 2**, a country i 's consumer chooses $\{c_{i1}, \dots, c_{in}\}$ by solving

$$\max_{\{c_{ij}\}_{j=1}^n} U_i = \left(\sum_{j=1}^n \chi_j^{\frac{1}{\varrho}} c_{ij}^{\frac{\varrho-1}{\varrho}} \right)^{\frac{\varrho}{\varrho-1}} \quad \text{subject to} \quad \sum_{j=1}^n \tau_{ij} c_{ij} = G_i, \quad (2.5)$$

where χ_j denotes a preference parameter, and G_i stands for the exogenously given budget of country i . Here, (2.5) is a standard CES aggregator with elasticity $\varrho > 1$.³

³We adopt a demand-side focus to highlight how network-leveraged trade costs shape flows, taking the production side as exogenous. Microfounding network dependence on the production side (e.g., extending Eaton and Kortum (2002)) would require an alternative specification (e.g., Lind and Ramondo, 2023); we leave this for future research. See Appendix 1.5.7 for related discussion.

The following conditions characterize the uniqueness of the nominal value of the optimal trade flow $\mu_{ij}^* = \tau_{ij}(\boldsymbol{\mu}^*)c_{ij}^*$, where c_{ij}^* denotes the solution to (2.5). For this purpose, define the network SAR operator

$$\mathbf{S} = I_N - \mathbf{A}, \quad \mathbf{A} = \lambda_d(I_n \otimes W) + \lambda_o(W \otimes I_n) + \lambda_w(W \otimes W),$$

with $\lambda_d = (\varrho - 1)\tilde{\lambda}_d$, $\lambda_o = (\varrho - 1)\tilde{\lambda}_o$, $\lambda_w = (\varrho - 1)\tilde{\lambda}_w$. Here, $I_n \otimes W$, $W \otimes I_n$, and $W \otimes W$ characterize, respectively, the cross-destination, cross-origin, and joint cross-origin/cross-destination linkages.

Assumption 2.4. (i) Let $b(x, y) = \lambda_d y + \lambda_o x + \lambda_w xy$ be a bilinear map for $x, y \in \mathbb{R}$. We assume

$$\max\{b(1, 1), b(1, \varphi_{\min}), b(\varphi_{\min}, 1), b(\varphi_{\min}, \varphi_{\min})\} < 1, \quad (2.6)$$

where $\varphi_{\min}$ is the minimum eigenvalue of W . Then, $\mathbf{S}$ is invertible. In addition, we assume⁴

$$\max\{|b(1, 1)|, |b(1, \varphi_{\min})|, |b(\varphi_{\min}, 1)|, |b(\varphi_{\min}, \varphi_{\min})|\} < 1, \quad (2.7)$$

then the Neumann series $\mathbf{S}^{-1} = \sum_{r=0}^{\infty} \mathbf{A}^r$ is well-defined.

(ii) $\boldsymbol{\mu}^*$ satisfies

$$\sup_{i,j} \sum_{k,l=1}^n \left| \sum_{p,q=1}^n s_{ij,pq} \left(\frac{\partial(\alpha_q(\boldsymbol{\mu}) + \eta_p(\boldsymbol{\mu}))}{\partial \ln(\mu_{kl})} \right) \right| < 1,$$

where $s_{ij,kl}$ denotes the $((j-1)n + i, (l-1)n + k)$ -element of $\mathbf{S}^{-1}$. Further,

$$\begin{aligned} \alpha_j(\boldsymbol{\mu}) &= -\frac{1}{2} \ln(G^W) + \ln(G_j) + \ln(\Pi_j^{\varrho-1}(\boldsymbol{\mu})), & j &= 1, \dots, n, \\ \eta_i(\boldsymbol{\mu}) &= -\frac{1}{2} \ln(G^W) + \ln(G_i) + \ln(P_i^{\varrho-1}(\boldsymbol{\mu})), & i &= 1, \dots, n, \end{aligned} \quad (2.8)$$

where

$$\Pi_j(\boldsymbol{\mu}) = \left(\sum_{i=1}^n \frac{G_i}{G^W} \left(\frac{\tau_{ij}(\boldsymbol{\mu})}{P_i(\boldsymbol{\mu})} \right)^{1-\varrho} \right)^{\frac{1}{1-\varrho}}, \quad P_i(\boldsymbol{\mu}) = \left(\sum_{j=1}^n \frac{G_j}{G^W} \left(\frac{\tau_{ij}(\boldsymbol{\mu})}{\Pi_j(\boldsymbol{\mu})} \right)^{1-\varrho} \right)^{\frac{1}{1-\varrho}}, \quad G^W = \sum_{i=1}^n G_i.$$

Our model's equilibrium is governed by two contraction mechanisms: one induced by the network SAR operator and the other by the conventional multilateral-resistance mapping.

⁴Under our maintained structure for W , this condition is equivalent to $\rho_{\text{spec}}(\mathbf{A}) < 1$, where $\rho_{\text{spec}}(\mathbf{A})$ denotes the spectral radius of $\mathbf{A}$.

Under conditions (i) and (ii) in Assumption 2.4, there exists a unique $\boldsymbol{\mu}^*$ satisfying

$$\mu_{ij}^* = \frac{G_i G_j}{G^W} \left(\frac{\tau_{ij}(\boldsymbol{\mu}^*)}{P_i(\boldsymbol{\mu}^*) \Pi_j(\boldsymbol{\mu}^*)} \right)^{1-\varrho} = \exp \left(\sum_{k,l=1}^n s_{ij,kl} (x'_{kl} \beta + \alpha_l(\boldsymbol{\mu}^*) + \eta_k(\boldsymbol{\mu}^*)) \right), \quad (2.9)$$

where $x_{ij} = (\ln(D_{ij,1}), \dots, \ln(D_{ij,K}))'$ and $\beta = (\beta_1, \dots, \beta_K)'$ with $\beta_k = (1-\varrho)\tilde{\beta}_k$. Under the no-information-friction setting, the expected flows coincide with the optimized flows, i.e., $\boldsymbol{\mu}^*$. Equation (2.9) is a semi-reduced form and establishes the main econometric equation by regarding $\alpha_j(\boldsymbol{\mu}^*)$ and $\eta_k(\boldsymbol{\mu}^*)$ as fixed-effect components. It is not a (full) reduced-form, however, since these structural fixed effects still depend on $\boldsymbol{\mu}^*$. See Appendix 1.3.2 for technical details.

As a result, the equilibrium trade cost between i and j ($i \neq j$) admits the decomposition

$$\ln \tau_{ij}(\boldsymbol{\mu}^*) = \underbrace{\sum_{k,l=1}^n s_{ij,kl} \ln \tau_{kl}^+}_{\text{gross propagated primitive-friction load}} - \underbrace{\frac{1}{\varrho-1} \sum_{k,l=1}^n \tilde{s}_{ij,kl} (\alpha_l(\boldsymbol{\mu}^*) + \eta_k(\boldsymbol{\mu}^*))}_{\text{propagated equilibrium correction}},$$

where $\tilde{s}_{ij,kl} = s_{ij,kl} - \mathbb{I}\{ij = kl\} = (e'_{nj} \otimes e'_{ni}) (\sum_{r=1}^{\infty} \mathbf{A}^r) (e_{nl} \otimes e_{nk})$ captures purely indirect propagation from kl to ij , excluding the direct own-pair component. The first block is the gross primitive-friction environment to which ij is exposed once bilateral geographic and institutional frictions are propagated through cross-destination, cross-origin, and hub-based linkages. The second block is an offsetting equilibrium correction generated by the indirect propagation of the structural origin- and destination-side terms.

Using $\alpha_l(\boldsymbol{\mu}^*) + \eta_k(\boldsymbol{\mu}^*) = \ln\left(\frac{G_k G_l}{G^W}\right) + (\varrho-1) \ln(P_k(\boldsymbol{\mu}^*) \Pi_l(\boldsymbol{\mu}^*))$, the second block can be further decomposed into a market-size adjustment and a multilateral-resistance adjustment. Hence, equilibrium bilateral costs are not merely pair-specific objects: they reflect both propagated primitive bilateral frictions and an offsetting multilateral-resistance correction through connected countries' market sizes and market-access conditions. Network leverage lowers the effective bilateral cost only when this equilibrium correction is sufficiently strong to outweigh the propagated primitive-friction load. When $\lambda_d = \lambda_o = \lambda_w = 0$, the network multiplier collapses to the identity, $\tilde{s}_{ij,kl} = 0$ for all ij, kl , the correction term vanishes, and the conventional iceberg-cost gravity model is recovered.

2.2 Econometric model specification

This subsection introduces our econometric model based on the established theoretical foundation. The true data-generating process (DGP) is specified as

$$y_{ij} = \mu_{ij}^0 \xi_{ij}, \quad \mu_{ij}^0 = \exp \left(\sum_{k,l=1}^n s_{ij,kl} (x'_{kl} \beta^0 + \alpha_l^0 + \eta_k^0) \right), \quad (2.10)$$

where $\mu_{ij}^0 = \mathbb{E}(y_{ij} \mid \mathbf{x})$; $\mathbf{x} = (x'_{11}, \dots, x'_{n1}, \dots, x'_{1n}, \dots, x'_{nn})'$ denotes the vector of exogenous characteristics; ξ_{ij} is a multiplicative CEF disturbance satisfying $\mathbb{E}(\xi_{ij} \mid \mathbf{x}) = 1$; $\lambda^0 = (\lambda_d^0, \lambda_o^0, \lambda_w^0)'$ denotes the vector of true network-spillover parameters; $\beta^0 = (\beta_1^0, \dots, \beta_K^0)'$ is the true coefficient vector on the covariates; and α_j^0 and η_i^0 represent the true origin- and destination-specific fixed effects, respectively. The main advantage of (2.10) is that estimation only requires the conditional mean restriction $\mathbb{E}(\xi_{ij} \mid \mathbf{x}) = 1$. For analytic convenience, define

$$u_{ij} = \mu_{ij}^0 (\xi_{ij} - 1), \text{ so that } y_{ij} = \mu_{ij}^0 + u_{ij}, \quad \mathbb{E}(u_{ij} \mid \mathbf{x}) = 0.$$

This mean-based formulation motivates our level specification and, in turn, the use of PPML estimation below.

Equation (2.10) addresses three essential issues. First, (2.10) directly specifies the economically relevant object $\mathbb{E}(y_{ij} \mid \mathbf{x})$. By contrast, the logarithmic transformation approach, as in LeSage and Pace (2008), encounters several well-known difficulties. As emphasized by Santos Silva and Tenreiro (2006), applying a logarithmic transformation in constant-elasticity models can lead to inconsistent estimation because it specifies $\mathbb{E}(\ln y_{ij} \mid \mathbf{x})$ rather than $\mathbb{E}(y_{ij} \mid \mathbf{x})$. By Jensen's inequality, $\mathbb{E}(\ln y_{ij} \mid \mathbf{x}) \neq \ln(\mathbb{E}(y_{ij} \mid \mathbf{x}))$, and the gap is $\mathbb{E}(\ln y_{ij} \mid \mathbf{x}) - \ln(\mathbb{E}(y_{ij} \mid \mathbf{x})) = \mathbb{E}(\ln \xi_{ij} \mid \mathbf{x})$. The resulting bias can be particularly severe when some y_{ij} values are very large. When the sample contains zero OD flows, an additional problem arises: $\ln(y_{ij})$ is not well defined, so the logarithmic specification itself is not feasible without an ad hoc transformation. A more detailed discussion is provided in Appendix 1.⁵

Second, in spatial econometric models the implied conditional mean of the level variable may depend directly on the assumed distribution of the transformed disturbance. Let v_{ij} denote the additive error term in the logarithmically transformed model of LeSage and Pace

⁵In addition to the issue $\mathbb{E}(\ln \xi_{ij} \mid \mathbf{x}) \neq 0$, various dependence structures between ξ_{ij} and components of $\mathbf{x}$ can also affect the magnitude of the bias from the log-transformed model. Since $\ln(\xi_{ij})$ admits a Maclaurin-series expansion, $\mathbb{E}(\ln \xi_{ij} \mid \mathbf{x})$ depends on the infinite sequence of conditional moments $h_p(\mathbf{x}) \equiv \mathbb{E}((\xi_{ij}^-)^p \mid \mathbf{x})$ for $p = 2, 3, \dots$, where $\xi_{ij}^- = \xi_{ij} - 1$. Hence, moments such as $\mathbb{E}(x_{ij} \ln \xi_{ij})$ depend on the correlation between x_{ij} and these higher-order conditional moments.

(2008). Assuming $y_{ij} > 0$ for all ij , their specification can be rewritten as

$$y_{ij} = \exp \left(\sum_{k,l=1}^n s_{ij,kl} x'_{kl} \beta \right) \prod_{k,l=1}^n \exp(v_{kl})^{s_{ij,kl}}.$$

Thus, $\mathbb{E}(y_{ij} \mid \mathbf{x})$ depends on the distribution of $\{v_{kl}\}$. For example, if $v_{ij} \mid \mathbf{x} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}(y_{ij} \mid \mathbf{x}) = \exp \left(\sum_{k,l=1}^n s_{ij,kl} x'_{kl} \beta \right) \exp \left(\frac{\sigma^2}{2} \sum_{k,l=1}^n s_{ij,kl}^2 \right),$$

since $\mathbb{E}(\exp(v_{kl})^{s_{ij,kl}} \mid \mathbf{x}) = \exp(\sigma^2 s_{ij,kl}^2 / 2)$. By contrast, if $v_{ij} \mid \mathbf{x} \stackrel{i.i.d.}{\sim} \text{logGamma}(\theta_{\text{shape}}, \theta_{\text{rate}})$ with $\theta_{\text{shape}} / \theta_{\text{rate}} = 1$, then

$$\mathbb{E}(y_{ij} \mid \mathbf{x}) = \exp \left(\sum_{k,l=1}^n s_{ij,kl} x'_{kl} \beta \right) \frac{\theta_{\text{shape}}^{\sum_{k,l=1}^n s_{ij,kl}} \prod_{k,l=1}^n \Gamma(\theta_{\text{shape}} + s_{ij,kl})}{\Gamma(\theta_{\text{shape}})^{n^2}}.$$

Hence, the level conditional mean implied by the logarithmic model of LeSage and Pace (2008) varies with the distributional specification imposed on $\{v_{kl}\}$.

Third, ad hoc transformations of y_{ij} become necessary because a logarithmically transformed specification only accommodates strictly positive outcomes. Since $\ln(0)$ is undefined, practitioners often replace y_{ij} by $y_{ij} + c$ with some $c > 0$, commonly $c = 1$. However, as emphasized by Chen and Roth (2024) and Mullahy and Norton (2024), the choice of c lacks a theoretical foundation and changes the object being modeled, namely $\mathbb{E}(\ln(y_{ij} + c) \mid \mathbf{x})$. This can lead to biased estimates and misleading interpretations of marginal effects. In the supplement, we show that the resulting distortion becomes more pronounced when the distribution of y_{ij} places more mass near zero.

2.3 Estimation

This subsection describes an estimation method for the parameters in (2.10), $\boldsymbol{\theta}^0 = (\theta^{0'}, \boldsymbol{\phi}^{0'})'$, $\theta^0 = (\lambda^{0'}, \beta^{0'})'$, $\boldsymbol{\phi}^0 = (\boldsymbol{\alpha}^{0'}, \boldsymbol{\eta}^{0'})'$, where $\boldsymbol{\alpha}^0 = (\alpha_1^0, \dots, \alpha_n^0)'$ and $\boldsymbol{\eta}^0 = (\eta_1^0, \dots, \eta_n^0)'$. For possible values of the parameters, we denote $\boldsymbol{\theta} = (\theta', \boldsymbol{\phi}')'$, $\theta = (\lambda', \beta')'$ and $\boldsymbol{\phi} = (\boldsymbol{\alpha}', \boldsymbol{\eta}')'$, where $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)'$ and $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)'$.

Assumption 2.5. Let Λ be the parameter space of λ . For each $\lambda \in \Lambda$, we define

$$\mathbf{A}(\lambda) = \lambda_d(I_n \otimes W) + \lambda_o(W \otimes I_n) + \lambda_w(W \otimes W) \text{ and } \mathbf{A} = \mathbf{A}(\lambda^0).$$

We assume $\sup_n \sup_{\lambda \in \Lambda} \|\mathbf{A}(\lambda)\|_\infty < 1$.

Assumption 2.5 guarantees the existence of the semi-reduced form at each $\lambda \in \Lambda$.⁶

The Poisson pseudo maximum likelihood (PPML) estimation method. We utilize the Poisson pseudo maximum likelihood (PPML) estimator developed by (Gourieroux et al., 1984; Santos Silva and Tenreiro, 2006). The log-likelihood function for $\boldsymbol{\theta}$ is:

$$\ell_N(\boldsymbol{\theta}) = \sum_{i,j=1}^n \ell_{ij}(\boldsymbol{\theta}) - \frac{1}{2} \left(\sum_{j=1}^n \alpha_j - \sum_{i=1}^n \eta_i \right)^2, \quad (2.11)$$

where $\ell_{ij}(\boldsymbol{\theta}) = -\mu_{ij}(\boldsymbol{\theta}) + y_{ij} \ln(\mu_{ij}(\boldsymbol{\theta})) - \ln(y_{ij}!)$ and the second term in (2.11) denotes a penalty term for the normalization (see Sec.2 in Fernandez-Val and Weidner (2016)).⁷ This method focuses on correctly specifying $\mathbb{E}(y_{ij}|\mathbf{x})$ without transforming the model.⁸ As the most notable advantage, hence, no additional assumptions are required on ξ_{ij} in estimation except $\mathbb{E}(\xi_{ij}|\mathbf{x}) = 1$. In conventional spatial econometric literature, the normal-based likelihood is employed. Since this conventional method uses the y 's conditional mean as well as its variance structure, it is vulnerable to unknown heteroskedasticity (Lin and Lee, 2010). On the other hand, since this method does not specify a specific variance structure, we can not only achieve consistency of the PPMLE but it can also accommodate a robust standard error for heteroskedasticity and autocorrelation (see Sec. 3.2).

Then, the PPML estimator (PPMLE) can be obtained by

$$\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}) = \arg \max_{\boldsymbol{\theta} \in \Theta_{\boldsymbol{\theta}}, \boldsymbol{\phi} \in \mathbb{R}^{2n}} \ell_N(\boldsymbol{\theta}, \boldsymbol{\phi}),$$

where $\Theta_{\boldsymbol{\theta}}$ is a parameter space of $\boldsymbol{\theta}$. For further analysis, let $\hat{\boldsymbol{\phi}}(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\phi} \in \mathbb{R}^{2n}} \ell_N(\boldsymbol{\theta}, \boldsymbol{\phi})$ for each $\boldsymbol{\theta} \in \Theta_{\boldsymbol{\theta}}$, and $\ell_N^c(\boldsymbol{\theta}) = \ell_N(\boldsymbol{\theta}, \hat{\boldsymbol{\phi}}(\boldsymbol{\theta}))$ denote the concentrated penalized log-likelihood function. For each element in $\hat{\boldsymbol{\phi}}(\boldsymbol{\theta})$, we define $\hat{\boldsymbol{\alpha}}(\boldsymbol{\theta}) = (\hat{\alpha}_1(\boldsymbol{\theta}), \dots, \hat{\alpha}_n(\boldsymbol{\theta}))'$ and $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}) =$

⁶This condition is slightly stronger than condition 2.6 in Assumption 2.4 (i). The purpose of this assumption is for asymptotic analysis.

⁷The key insight of (2.11) is that we can work with an unconstrained optimization problem. Indeed, (2.11) imposes a single linear constraint, $v' \boldsymbol{\phi} = 0$ where $v = (l'_n, -l'_n)'$, to eliminate the identification issue originated from the fixed effects' additive feature: $\alpha_j + \eta_i = \alpha_j^* + \eta_i^*$ where $\alpha_j^* = \alpha_j + c$ and $\eta_i^* = \eta_i - c$ for any c . Note that a normalization restriction for this issue is not unique: Fernandez-Val and Weidner (2016) additionally mention the possibility of $\alpha_1 = 0$, while Lee and Yu (2010) employs $\sum_{i=1}^n \eta_i = 0$.

⁸Another approach for estimating $\mathbb{E}(y_{ij}|\mathbf{x})$ without transforming the model is employing non-linear least squares (NLS). However, this method tends to give disproportionate weight to noisy observations, leading to inefficient estimations. This inefficiency arises because the method heavily depends on a relatively small number of observations (Silva and Tenreiro (2006, Sec. III A)). For details, refer to Section 2.1 in the supplement.

$(\hat{\eta}_1(\theta), \dots, \hat{\eta}_n(\theta))'$ for each $\theta \in \Theta_\theta$.

Efficient computation of the network multiplier matrix $\mathbf{S}^{-1}$. Computation of our model is more complex than that of conventional gravity models, as it involves two contraction mappings. In estimation, however, inverting $\mathbf{S}$ is sufficient since it leads to the semi-reduced-form (2.9). In detail, we encounter the evaluation of $\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta})$ for each $\boldsymbol{\theta}$, where $\mathbf{Z}(\boldsymbol{\theta}) = \mathbf{X}\beta + \boldsymbol{\alpha} \otimes l_n + l_n \otimes \boldsymbol{\eta}$ with $\mathbf{Z} = \mathbf{Z}(\boldsymbol{\theta}^0)$. This term must be carried out repeatedly during likelihood maximization. As n increases, the computation cost grows very rapidly with n . By exploiting the structure of the three networks $I_n \otimes W$, $W \otimes I_n$, and $W \otimes W$ generated by a single row-normalized connectivity matrix W , we can design a much more efficient computation procedure.

By Assumption 2.2 and the spectral decomposition theorem, $W = QDQ^{-1}$, where $D = \text{diag}(\varphi_1, \dots, \varphi_n)$ and Q denotes the eigenvector matrix. Because W is constructed from symmetric relationships, all φ_i are real. Since $I_n \otimes W$, $W \otimes I_n$, and $W \otimes W$ share the same eigenvector basis, an eigenvalue of $\mathbf{A}(\lambda)$ can be represented by

$$\mathbf{A}(\lambda)(q_i \otimes q_j) = (\lambda_d \varphi_j + \lambda_o \varphi_i + \lambda_w \varphi_i \varphi_j)(q_i \otimes q_j) \text{ for } i, j = 1, \dots, n, \quad (2.12)$$

where q_i denotes the i th column vector of Q .⁹

Define $Z^{\text{mat}}(\boldsymbol{\theta})$ satisfying $\mathbf{Z}(\boldsymbol{\theta}) = \text{vec}(Z^{\text{mat}}(\boldsymbol{\theta}))$. Then, we want to obtain the matrix fixed-point $T(\boldsymbol{\theta})$ satisfying $Z^{\text{mat}}(\boldsymbol{\theta}) = T(\boldsymbol{\theta}) - \lambda_d WT(\boldsymbol{\theta}) - \lambda_o T(\boldsymbol{\theta})W' - \lambda_w WT(\boldsymbol{\theta})W'$ ($\Leftrightarrow \text{vec}(T(\boldsymbol{\theta})) = \mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta})$). Note that Q and D are invariant in the estimation procedure. Hence, we have

$$\tilde{Z}^{\text{mat}}(\boldsymbol{\theta}) = \tilde{T}(\boldsymbol{\theta}) - \lambda_d D \tilde{T}(\boldsymbol{\theta}) - \lambda_o \tilde{T}(\boldsymbol{\theta}) D - \lambda_w D \tilde{T}(\boldsymbol{\theta}) D,$$

where $\tilde{Z}^{\text{mat}}(\boldsymbol{\theta}) = Q^{-1}Z^{\text{mat}}(\boldsymbol{\theta})Q^{-1'}$ and $\tilde{T}(\boldsymbol{\theta}) = Q^{-1}T(\boldsymbol{\theta})Q^{-1'}$. It implies

$$(\tilde{T}(\boldsymbol{\theta}))_{ij} = \frac{(\tilde{Z}^{\text{mat}}(\boldsymbol{\theta}))_{ij}}{1 - \lambda_d \varphi_i - \lambda_o \varphi_j - \lambda_w \varphi_i \varphi_j} \text{ for } i, j = 1, \dots, n, \quad (2.13)$$

by (2.12). Then, we can easily recover $T(\boldsymbol{\theta}) = Q\tilde{T}(\boldsymbol{\theta})Q'$.

⁹In terms of model structure, we can allow different proximity matrices for cross-origin and cross-destination linkages (W and M) such as Jeong and Lee (2024). In this case, however, it might not be possible to dramatically reduce computation costs since two connectivity matrices do not share the eigenvector basis in general. We leave this issue for future research.

Identification condition. Next, we study the identification of θ^0 . We define the following notations for this purpose:

Let $\mathbf{H}^{\theta\theta}(\theta) = \begin{bmatrix} \mathbf{H}^{\lambda\lambda}(\theta) & \mathbf{H}^{\beta\lambda'}(\theta) \\ \mathbf{H}^{\beta\lambda}(\theta) & \mathbf{H}^{\beta\beta}(\theta) \end{bmatrix}$ with $\mathbf{H}^{\lambda\lambda}(\theta) = (h_{ab}^{\lambda\lambda}(\theta))$ ($a, b \in \{d, o, w\}$), $\mathbf{H}^{\beta\lambda}(\theta) = [h_d^{\beta\lambda}(\theta) \ h_o^{\beta\lambda}(\theta) \ h_w^{\beta\lambda}(\theta)]$, $\mathbf{H}^{\beta\beta}(\theta) = \mathbf{0}_{K \times K}$, $\mathbf{H}^{\phi\theta}(\theta) = [h_d^{\phi\theta}(\theta) \ h_o^{\phi\theta}(\theta) \ h_w^{\phi\theta}(\theta) \ \mathbf{0}_{2n \times K}]$, and $\mathbf{H}^{\phi\phi} = - \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \otimes l_n l_n'$. Here,

- $h_{ab}^{\lambda\lambda}(\theta) = (2\mathbf{W}_a \mathbf{W}_b \mathbf{S}^{-3}(\lambda) \mathbf{Z}(\theta))' \mathbf{u}(\theta)$ for $a, b \in \{d, o, w\}$, $h_a^{\beta\lambda}(\theta) = (\mathbf{W}_a \mathbf{S}^{-2}(\lambda) \mathbf{X})' \mathbf{u}(\theta)$ and $h_a^{\phi\theta}(\theta) = (\mathbf{W}_a \mathbf{S}^{-2}(\lambda) \mathbf{D})' \mathbf{u}(\theta)$ for $a \in \{d, o, w\}$, where $\mathbf{W}_d = I_n \otimes W$, $\mathbf{W}_o = W \otimes I_n$, and $\mathbf{W}_w = W \otimes W$;
- $\mathbf{X}$ denotes an $N \times K$ matrix which has $x_{ij,k}$ as the $((j-1)n + i, k)$ - element of $\mathbf{X}$; $\mathbf{D} = [I_n \otimes l_n, l_n \otimes I_n]$ is an $N \times 2n$ matrix for dummy variables;
- $\boldsymbol{\mu}(\theta) = (\exp(\tilde{\mu}_{11}(\theta)), \dots, \exp(\tilde{\mu}_{n1}(\theta)), \dots, \exp(\tilde{\mu}_{1n}(\theta)), \dots, \exp(\tilde{\mu}_{nn}(\theta)))$ with $\boldsymbol{\mu}^0 = \boldsymbol{\mu}(\theta^0)$ and $\tilde{\mu}_{ij}(\theta) = \sum_{k,l=1}^n s_{ij,kl}(\lambda) (x'_{kl}\beta + \alpha_l + \eta_k)$;
- $\mathbf{u}(\theta) = \mathbf{y} - \boldsymbol{\mu}(\theta)$ with $\mathbf{u} = \mathbf{y} - \boldsymbol{\mu}^0$.

Assumption 2.6 (Identification). Let $\Theta = \Theta_\theta \times \Phi$ be the parameter space of θ , where Θ_θ denotes a compact parameter space of θ and Φ represents a parameter space of ϕ . Here, $\Phi \subset [-C, C]^{2n}$ for some finite constant $C > 0$.

(i) For each $(\theta, \phi) \in \Theta$, define $\mathbf{J}_N^{\phi\phi}(\theta) = \frac{1}{N} (\mathbf{D}' \mathbf{S}^{-1'}(\lambda) \text{Diag}(\boldsymbol{\mu}(\theta)) \mathbf{S}^{-1}(\lambda) \mathbf{D} - \mathbf{H}^{\phi\phi})$. Assume $\liminf_{n \rightarrow \infty} \inf_{\theta \in \Theta_\theta} \inf_{\phi \in \Phi} \varphi_{\min}(\mathbf{J}_N^{\phi\phi}(\theta, \phi)) > 0$. Then, for each $\theta \in \Theta_\theta$ and for n sufficiently large, $\hat{\phi}(\theta) = \arg \max_{\phi \in \Phi} \ell_N(\theta, \phi)$ is unique.

(ii) For each $(\theta, \phi) \in \Theta$, define

$$\mathbf{J}_N^{\theta\theta}(\theta) = \frac{1}{N} (\mathbf{G}(\theta)' \mathbf{S}^{-1'}(\lambda) \text{Diag}(\boldsymbol{\mu}(\theta)) \mathbf{S}^{-1}(\lambda) \mathbf{G}(\theta) - \mathbf{H}^{\theta\theta}(\theta)),$$

$$\mathbf{J}_N^{\theta\phi}(\theta) = \frac{1}{N} (\mathbf{G}(\theta)' \mathbf{S}^{-1'}(\lambda) \text{Diag}(\boldsymbol{\mu}(\theta)) \mathbf{S}^{-1}(\lambda) \mathbf{D} - \mathbf{H}^{\phi\theta}(\theta)'), \text{ and } \mathbf{J}_N^{\phi\theta}(\theta) = (\mathbf{J}_N^{\theta\phi}(\theta))'.$$

Here, $\mathbf{G}(\theta) = [\mathbf{W}_d \mathbf{S}^{-1}(\lambda) \mathbf{Z}(\theta), \mathbf{W}_o \mathbf{S}^{-1}(\lambda) \mathbf{Z}(\theta), \mathbf{W}_w \mathbf{S}^{-1}(\lambda) \mathbf{Z}(\theta), \mathbf{X}]$. For each $\theta \in \Theta_\theta$, let

$$\hat{\mathbf{J}}_N^{\theta\theta}(\theta) = \mathbf{J}_N^{\theta\theta}(\theta, \hat{\phi}(\theta)), \quad \hat{\mathbf{J}}_N^{\theta\phi}(\theta) = \mathbf{J}_N^{\theta\phi}(\theta, \hat{\phi}(\theta)), \quad \hat{\mathbf{J}}_N^{\phi\phi}(\theta) = \mathbf{J}_N^{\phi\phi}(\theta, \hat{\phi}(\theta)).$$

Assume $\liminf_{n \rightarrow \infty} \inf_{\theta \in \Theta_\theta} \varphi_{\min}(\hat{\mathbf{H}}(\theta)) > 0$, where $\hat{\mathbf{H}}(\theta) = \hat{\mathbf{J}}_N^{\theta\theta}(\theta) - \hat{\mathbf{J}}_N^{\theta\phi}(\theta) [\hat{\mathbf{J}}_N^{\phi\phi}(\theta)]^{-1} \hat{\mathbf{J}}_N^{\phi\theta}(\theta)$.

Then, for n sufficiently large, $\hat{\theta} = \arg \max_{\theta \in \Theta_\theta} \ell_N^c(\theta)$ is unique.

Assumption 2.6 (i) guarantees that, for each $\theta \in \Theta_\theta$, the population criterion $\ell_\infty(\theta, \phi)$ is strictly concave in ϕ , so that the maximizer $\phi(\theta)$ is unique. Assumption 2.6 (ii) ensures that the profiled criterion $\ell_\infty^c(\theta) = \ell_\infty(\theta, \phi(\theta))$ is strictly concave in θ , which implies that the θ^0 is the unique maximizer of $\ell_\infty^c(\theta)$. Consequently, (θ^0, ϕ^0) is point-identified in large samples, where $\phi^0 = \phi(\theta^0)$. The matrices $\mathbf{H}^{\theta\theta}(\theta)$, $\mathbf{H}^{\phi\theta}(\theta)$, and $\mathbf{H}^{\phi\phi}(\theta)$ are additional components in the Hessian that arise from the model's network interactions. Unlike the traditional gravity equation, the presence of spatial spillovers makes the curvature of the objective function more complex. Intuitively, when $\mathbf{S}^{-1}(\lambda)$ is well defined, the leading part of $\widehat{\mathbf{H}}(\theta)$ is likely positive definite; if the perturbation induced by the $\mathbf{H}$ -terms is not too large, the conditions in Assumption 2.6 are satisfied. Motivated by this identification argument, we recommend first estimating the conventional gravity equation to obtain preliminary estimates $\tilde{\beta}$, $\tilde{\alpha}$, and $\tilde{\eta}$. One can then set the initial parameter vector for maximizing (2.11) as $\hat{\theta}^{(0)} = (0, 0, 0, \tilde{\beta}', \tilde{\alpha}', \tilde{\eta}')'$, since $\mathbf{S}^{-1}(\lambda) = I_N$ when $\lambda = \mathbf{0}$. Details are provided in Lemmas 2.4 and 2.5 of Sec. 2.3 of the supplement file.

Note that the minimum eigenvalue of W ($\varphi_{\min}$) is a key network statistic showing bipartiteness (if $\varphi_{\min} \rightarrow -1$) and averaging rate (if $\varphi_{\min} \rightarrow 0$) (see Chung (1997); Bramoullé et al. (2014) for more details).

3 Statistical Analysis

This section derives the asymptotic distribution of the PPMLE, presents relevant statistical inference, and presents simulation results for finite samples.

3.1 Asymptotic distribution of the PPMLE

To derive the asymptotic distribution of $\hat{\theta}$ and $\hat{\phi}$, we impose the following regularity assumption for Theorems 3.1 and 3.2. Details can be found in Sec. 2 of the supplement file.

Assumption 3.1. (i) $\{x_{ij}\}$, $\{\eta_i^0\}$, and $\{\alpha_j^0\}$ are random fields satisfying $\max_k \sup_{i,j,n} |x_{ij,k}| < C$, $\sup_{i,n} |\eta_i^0| < C$, and $\sup_{j,n} |\alpha_j^0| < C$, where $C > 0$ denotes a generic finite constant.

(ii) $\{\xi_{ij}\}$ is a random field satisfying $\sup_{i,j,n} \mathbb{E}|\xi_{ij}|^{2+c} < C$ for some $c > 0$.

(iii) $\mathbb{E}(\xi_{ij}|\mathbf{x}) = 1$ for all $i, j = 1, \dots, n$.

Then, the same type of properties follow for the additive error u_{ij} : $\mathbb{E}(u_{ij}|\mathbf{x}) = 0$ and $\sup_{i,j,n} \mathbb{E}|u_{ij}|^{2+c} < C$. The theorems below state the asymptotic properties of the PPMLE. The asymptotic properties of the PPMLE for the fixed-effect parameters are used to examine the structure of the multilateral resistance terms. Details are in Sec. 2.3 of the supplement.

Theorem 3.1. Suppose that Assumptions 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 3.1, and 3.2 hold. Let

$$\Sigma_{\theta,N} = \frac{1}{N} \mathbf{G}' \mathbf{S}^{-1'} \text{Diag}(\boldsymbol{\mu}) \mathbf{M}_D \mathbf{S}^{-1} \mathbf{G}, \text{ and } \Omega_{\theta,N} = \frac{1}{N} \mathbf{G}' \mathbf{S}^{-1'} \mathbf{M}_D' \mathbb{E}(\mathbf{u}\mathbf{u}') \mathbf{M}_D \mathbf{S}^{-1} \mathbf{G},$$

where $\mathbf{M}_D = I_N - \mathbf{P}_D \text{Diag}(\boldsymbol{\mu})$ with $\mathbf{P}_D = \mathbf{S}^{-1} \mathbf{D} (\widetilde{\mathbf{D}'\mathbf{D}})^{-1} \mathbf{D}' \mathbf{S}^{-1'}$ and $\widetilde{\mathbf{D}'\mathbf{D}} = \mathbf{D}' \mathbf{S}^{-1'} \text{Diag}(\boldsymbol{\mu}) \mathbf{S}^{-1} \mathbf{D} - \mathbf{H}^{\phi\phi} = N \mathbf{J}_N^{\phi\phi}(\boldsymbol{\theta}^0)$.

Suppose (i) $\liminf_{n \rightarrow \infty} \varphi_{\min}(\Omega_{\theta,N}) > 0$ and (ii) $\Sigma_{\theta} = \text{plim}_{n \rightarrow \infty} \Sigma_{\theta,N}$ exists and is positive definite. Then, we have

$$\sqrt{N} (\hat{\theta} - \theta^0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_{\theta}^{-1} \Omega_{\theta} \Sigma_{\theta}^{-1}) \text{ as } n \rightarrow \infty, \quad (3.1)$$

where $\Omega_{\theta} = \text{plim}_{n \rightarrow \infty} \Omega_{\theta,N}$.

Theorem 3.2. Suppose that Assumptions 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 3.1 and 3.2 hold. Let

$$\mathbf{V}_{\phi,N} = n (\widetilde{\mathbf{D}'\mathbf{D}})^{-1} \mathbf{D}' \mathbf{S}^{-1'} \mathbf{M}_{\phi}' \mathbb{E}(\mathbf{u}\mathbf{u}') \mathbf{M}_{\phi} \mathbf{S}^{-1} \mathbf{D} (\widetilde{\mathbf{D}'\mathbf{D}})^{-1},$$

where $\mathbf{M}_{\phi} = I_N - \mathbf{M}_D \mathbf{S}^{-1} \mathbf{G} (\mathbf{G}' \mathbf{S}^{-1'} \text{Diag}(\boldsymbol{\mu}) \mathbf{M}_D \mathbf{S}^{-1} \mathbf{G})^{-1} \mathbf{G}' \mathbf{S}^{-1'} \text{Diag}(\boldsymbol{\mu})$. Then,

$$\begin{aligned} \sqrt{n} (\hat{\alpha}_j - \alpha_j^0) &\xrightarrow{d} \mathcal{N}(0, \lim_{n \rightarrow \infty} e_{2n,j}' \mathbf{V}_{\phi,N} e_{2n,j}), \text{ and} \\ \sqrt{n} (\hat{\eta}_i - \eta_i^0) &\xrightarrow{d} \mathcal{N}(0, \lim_{n \rightarrow \infty} e_{2n,n+i}' \mathbf{V}_{\phi,N} e_{2n,n+i}) \end{aligned}$$

as $n \rightarrow \infty$, where $e_{2n,j}$ denotes the $2n$ -dimensional unit vector with its j -th element equal to 1 and all other elements equal to 0.

By Assumption 2.6(i), we obtain both a well-defined $\hat{\phi}(\theta)$ and good conditioning of $\widetilde{\mathbf{D}'\mathbf{D}}$, which in turn ensures that $\mathbf{P}_D$ and $\mathbf{M}_D$ are well-defined. Assumption 2.6 as a whole guarantees identification of θ^0 and, in particular, implies that $\text{plim}_{n \rightarrow \infty} \widehat{\mathbf{H}}(\theta^0) = \Sigma_{\theta}$, i.e., $\widehat{\mathbf{H}}(\theta^0)$ is asymptotically equivalent to $\Sigma_{\theta,N}$. Details are provided in Lemmas 2.4 and 2.5 of Sec. 2.3 of the supplement file.

3.2 Variance estimation

This subsection provides a method for spatial HAC estimation of the covariance matrices in Theorems 3.1 and 3.2 of the PPMLE. Because errors in bilateral flows may exhibit network dependence across origin–destination pairs like the network dependence embedded in the conditional mean, we conduct inference using spatial HAC covariance estimation without imposing a parametric dependence structure. Our suggested method extends the existing

literature on cross-sectional data to OD flows (Kelejian and Prucha (2007) and Kim and Sun (2011)), spatial extensions of the time-series literature (Newey and West (1987); Andrews (1991); de Jong and Davidson (2000)). Since an estimate of $\boldsymbol{\Sigma}_\theta$ is obtained by plugging $\hat{\boldsymbol{\theta}}$ in, we focus on estimating $\boldsymbol{\Omega}_\theta$ here. Also, note that estimating $\lim_{n \rightarrow \infty} e'_{2n,j} \mathbf{V}_{\phi,N} e_{2n,j}$ and $\lim_{n \rightarrow \infty} e'_{2n,n+i} \mathbf{V}_{\phi,N} e_{2n,n+i}$ for $i, j = 1, \dots, n$ is applicable by the same way of estimating $\boldsymbol{\Omega}_\theta$.

First, we provide the regularity assumptions.

Assumption 3.2. (i) For the structure of $\mathbf{u} = (u_{11}, \dots, u_{n1}, \dots, u_{1n}, \dots, u_{nn})'$, we assume

$$\mathbf{u} = \mathbf{B}\mathbf{H}\boldsymbol{\epsilon}, \quad (3.2)$$

where $\mathbf{B}$ denotes some $N \times N$ matrix, $\mathbf{H} = \text{diag}(\sigma_{11}^*, \dots, \sigma_{n1}^*, \dots, \sigma_{1n}^*, \dots, \sigma_{nn}^*)$, and $\boldsymbol{\epsilon} = (\epsilon_{11}, \dots, \epsilon_{n1}, \dots, \epsilon_{1n}, \dots, \epsilon_{nn})'$ is an $N \times 1$ vector of innovations.

(ii) $\epsilon_{ij} \stackrel{i.i.d.}{\sim} (0, 1)$ across ij with $\sup_{n,i,j} \mathbb{E}|\epsilon_{ij}|^4 < \infty$.

(iii) $0 < \inf_{i,j,n} \sigma_{ij}^* \leq \sup_{i,j,n} \sigma_{ij}^* < \infty$.

(iv) $\mathbf{B}$ is nonsingular and $\sup_n \max\{\|\mathbf{B}\|_\infty, \|\mathbf{B}\|_1\} < \infty$.

Assumption 3.2 (i) describes the basic covariance structure. Hence, we have

$$\boldsymbol{\Omega}_{\theta,N} = \frac{1}{N} \mathbf{R}' \mathbf{R} = \frac{1}{N} \sum_{i,j,k,l=1}^n R_{ij} R'_{kl} = \frac{1}{N} \sum_{i,j,k,l=1}^n \mathbb{E} \left(\left(\mathbf{G}' \mathbf{S}^{-1'} \mathbf{M}'_{\mathbf{D}} \mathbf{u} \right)_{.,ij} \left(\mathbf{u}' \mathbf{M}_{\mathbf{D}} \mathbf{S}^{-1} \mathbf{G} \right)_{kl,.} \right),$$

where $\mathbf{R} = \mathbf{H} \mathbf{B}' \mathbf{M}_{\mathbf{D}} \mathbf{S}^{-1} \mathbf{G}$, R_{ij} denotes the $((j-1)n+i)$ -th column of $\mathbf{R}$, $(\mathbf{G}' \mathbf{S}^{-1'} \mathbf{M}'_{\mathbf{D}} \mathbf{u})_{.,ij}$ denotes the $(j-1)n+i$ -th column of $\mathbf{G}' \mathbf{S}^{-1'} \mathbf{M}'_{\mathbf{D}} \mathbf{u}$ and $(\mathbf{u}' \mathbf{M}_{\mathbf{D}} \mathbf{S}^{-1} \mathbf{G})_{kl,.}$ is the $(l-1)n+k$ -th row of $\mathbf{u}' \mathbf{M}_{\mathbf{D}} \mathbf{S}^{-1} \mathbf{G}$. Sec. 3.3 provides an example of (3.2). Conditions (ii) and (iii) in Assumption 3.2 are conventional. Condition (iv) comes from Kelejian and Prucha (2007).¹⁰

Based on this setting, we can characterize the spatial HAC estimator

$$\hat{\boldsymbol{\Omega}}_{\theta,N} = \frac{1}{N} \sum_{i,j,k,l=1}^n \left(\widehat{\mathbf{G}}' \widehat{\mathbf{S}}^{-1'} \widehat{\mathbf{M}}'_{\mathbf{D}} \hat{\mathbf{u}} \right)_{.,ij} \left(\hat{\mathbf{u}}' \widehat{\mathbf{M}}_{\mathbf{D}} \widehat{\mathbf{S}}^{-1} \widehat{\mathbf{G}} \right)_{kl,.} \mathbf{K} \left(\frac{d_{ij,kl}^*}{d_N} \right),$$

where $\hat{\cdot}$ is the plug-in estimator from $\hat{\boldsymbol{\theta}}$, $\mathbf{K}(\cdot)$ denotes a real-valued kernel function (e.g., Bartlett, Parzen, and Tukey-Hanning), $d_{ij,kl}^*$ is a distance measure between two pairs ij and kl , and d_N is a bandwidth. For details, refer to Sec. 2.4 in the supplement file.

¹⁰Still, it can be relaxed by the ideas of Pesaran and Yang (2020, 2021). For example, if the error structure follows (3.3) in the simulation section, we can allow a finite number of moderate dominant units. We will leave this issue for future research.

3.3 Monte Carlo Simulations

Performance of the spectral algorithm for $\mathbf{S}^{-1}\mathbf{Z}(\boldsymbol{\theta})$. First, we study the performance of our suggested algorithm for evaluating $\mathbf{S}^{-1}\mathbf{Z}(\boldsymbol{\theta})$ introduced in Section 2.3. Table 1 reports how the computation time of the two methods varies with n (in seconds) for completing the maximization of the log-likelihood function (2.11). Method A maximizes the log-likelihood based on the direct computation of $\mathbf{S}^{-1}(\lambda)$, while Method B utilizes our algorithm. As n increases from 9 to 64, the running time of Method A rises dramatically from 0.57 seconds to about 9,466 seconds (approximately 2.6 hours), whereas Method B increases only modestly from 0.35 seconds to 9.28 seconds. Consequently, the ratio A/B grows sharply from 1.61 to about 1,020, indicating that Method B is only slightly faster for very small problems but becomes more than three orders of magnitude faster for $n = 64$.

Table 1: Computation time comparison

n	$N = n^2$	Method A	Method B	A/B
9	81	0.5668	0.3518	1.6112
25	625	12.3862	0.3967	31.2265
49	2401	1252.0077	3.9725	315.1687
64	4096	9465.7145	9.2816	1019.8380

Note: Each number in this table represents seconds for finishing maximization of the log-likelihood (2.11). For the maximization, we use `fminunc`. Method A is maximizing the log-likelihood function based on the calculation of $\mathbf{S}^{-1}$ and multiplication by $\mathbf{Z}(\boldsymbol{\theta})$. Method B is maximizing the log-likelihood function based on (2.13). In the last column, we provide the ratios of the computation times for the two methods (in seconds). All computations were performed in MATLAB R2023b on a Windows 11 machine equipped with a 13th Gen Intel Core i5-1340P CPU (12 cores, 1.9 GHz) and 16 GB of RAM.

Data generating process. We consider $n = 49$ regions. Following Pesaran and Yang (2020, 2021), the regional proximity matrix W is constructed to feature two “dominant” regions (units 1 and 2). For regions $i \in \{1, 2\}$, the code draws relatively large link weights $\sim \mathcal{U}[0.8, 1]$ to n^{δ_i} neighbors (with $\delta_1 = 0.25$, $\delta_2 = 0.1$), then spreads the remaining mass across two randomly chosen non-hub neighbors so that each row approximately sums to one. Self-links are zero. Let the total world GDP be $G^W = 1400$. Country i ’s GDP G_i is proportional to its out-degree in W : $G_i = G^W \cdot s_i$ and $s_i = \frac{\sum_j (w_{ij} + 0.01)}{\sum_{i,k=1}^n (w_{ik} + 0.01)}$. We impose balanced trade targets so that exports by origin and imports by destination both equal G . Preferences follow an Armington structure with elasticity of substitution $\varrho = 5$ (Anderson and van Wincoop, 2003). First-stage parameters $\tilde{\lambda} = (\tilde{\lambda}_d, \tilde{\lambda}_o, \tilde{\lambda}_w)' = (0.05, 0.05, 0.025)'$ and $\tilde{\beta} = (-0.15, -0.05)'$ are mapped to reduced-form coefficients $\lambda^0 = (\varrho - 1)\tilde{\lambda} = (0.2, 0.2, 0.1)'$ and $\beta^0 = (1 - \varrho)\tilde{\beta} = (0.6, 0.2)'$. Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2] \in \mathbb{R}^{N \times 2}$ be bilateral covariates drawn from

$\mathcal{U}[0, 0.75]$, and $X^U = (x_1^u, \dots, x_n^u)' \in \mathbb{R}^{n \times 1}$ a standardized country characteristic. Second-stage loadings are $\tilde{\gamma}_o = \tilde{\gamma}_d = 0.01$ with the same scaling $\gamma = (1 - \varrho)\tilde{\gamma} = -0.04$.

Starting with this setting, the initial destination and origin fixed effects are specified by

$$\boldsymbol{\alpha}^{(0)} = X^U \gamma_o, \text{ and } \boldsymbol{\eta}^{(0)} = X^U \gamma_d.$$

Conditional on $(\boldsymbol{\alpha}^{(0)}, \boldsymbol{\eta}^{(0)})$, $\boldsymbol{\mu}^{(0)}$. Given $\boldsymbol{\mu}^{(0)}$ is determined. Given $\boldsymbol{\mu}^{(0)}$ and initial $(P^{(0)}, \Pi^{(0)}) = (\mathbf{1}_n, \mathbf{1}_n)$, we compute multilateral resistances via contraction mappings by (2.8):

$$\boldsymbol{\alpha}^{(\ell)} = c_0 \mathbf{1}_n + X^U \gamma_o + \log G + (\varrho - 1) \log \Pi^{(\ell-1)} \text{ and } \boldsymbol{\eta}^{(\ell)} = c_0 \mathbf{1}_n + X^U \gamma_d + \log G + (\varrho - 1) \log P^{(\ell-1)},$$

for $\ell = 1, 2, \dots$, where $c_0 = -\frac{1}{2} \log G^W$ and $G = (G_1, \dots, G_n)'$. Then, we have $\boldsymbol{\mu}^{(\ell)}$ for $\ell = 1, 2, \dots$. Note that we apply the normalization $\sum_i \alpha_i^{(\ell)} - \sum_i \eta_i^{(\ell)} = 0$ for each iteration ℓ . We iterate until $\max\{\|\boldsymbol{\alpha}^{(\ell)} - \boldsymbol{\alpha}^{(\ell-1)}\|_\infty, \|\boldsymbol{\eta}^{(\ell)} - \boldsymbol{\eta}^{(\ell-1)}\|_\infty\} < 10^{-12}$. Condition in Assumption 2.4 is utilized for guaranteeing this convergence. For each simulation replication, we allow random variations in the fixed effects:

$$\boldsymbol{\alpha}^0 = \boldsymbol{\alpha}^{(\infty)} + \boldsymbol{\varepsilon}^\alpha, \text{ and } \boldsymbol{\eta}^0 = \boldsymbol{\eta}^{(\infty)} + \boldsymbol{\varepsilon}^\eta,$$

where $\boldsymbol{\varepsilon}^\alpha = (\varepsilon_1^\alpha, \dots, \varepsilon_n^\alpha) \sim \mathcal{N}(\mathbf{0}, 0.08^2 I_n)$ and $\boldsymbol{\varepsilon}^\eta = (\varepsilon_1^\eta, \dots, \varepsilon_n^\eta) \sim \mathcal{N}(\mathbf{0}, 0.08^2 I_n)$. Consequently, we have $\boldsymbol{\mu}^0 = \exp(\mathbf{S}^{-1}(\mathbf{X}\beta^0 + \boldsymbol{\alpha}^0 \otimes \mathbf{1}_n + \mathbf{1}_n \otimes \boldsymbol{\eta}^0))$.

Next, we generate the error components for each simulation replication:

1. First, we generate $\xi_{ij}^* \stackrel{i.i.d.}{\sim} \text{Lognormal}(-\frac{1}{2}\sigma^2, \sigma^2)$ across ij with $\sigma^2 = 0.125^2$. Then, $\mathbb{E}(\xi_{ij}^* | \mathbf{x}) = \mathbb{E}(\xi_{ij}^*) = 1$.
2. Let $\epsilon_{ij}^* = \mu_{ij}^0$ for all ij . Then, $\mathbb{E}(\epsilon_{ij}^* | \mathbf{x}) = \mu_{ij}^0 \cdot (\mathbb{E}(\xi_{ij}^* | \mathbf{x}) - 1) = 0$ and $\text{Var}(\epsilon_{ij}^* | \mathbf{x}) = (\mu_{ij}^0)^2 \cdot \text{Var}(\xi_{ij}^*) = (\mu_{ij}^0)^2 \cdot (\exp(\sigma^2) - 1)$.
3. Last, we generate

$$u_{ij} = 0.008 \sum_{k=1}^n w_{ik}^* \epsilon_{kj}^* + 0.008 \sum_{l=1}^n w_{jl}^* \epsilon_{li}^* + 0.002 \sum_{k,l=1}^n w_{ik}^* w_{jl}^* \epsilon_{kl}^*, \quad (3.3)$$

where $W^* = (w_{ij}^*)$ is a row-normalized one characterized by the adjacency based on W (i.e., $w_{ij}^* = \frac{w_{ij}^*}{\sum_{k=1}^n w_{ik}^*}$ where $\tilde{w}_{ij}^* = \mathbb{I}\{w_{ij} + w_{ji} > 0\}$). This error structure follows (3.2) since $\sigma_{ij}^* = \mu_{ij}^0 \sqrt{\exp(\sigma^2) - 1}$ and $\mathbf{B} = 0.008(I_n \otimes W^*) + 0.008(W^* \otimes I_n) + 0.002(W^* \otimes W^*)$.

Basic information. Four criteria are used to evaluate the finite sample performance of the PPMLE: (i) empirical bias, (ii) empirical standard deviation (STD), (iii) standard error (s.e.), and (iv) the coverage probability of a nominal 95% confidence interval (CP). To evaluate the standard errors, we consider three kernel functions: (i) Bartlett, (ii) Parzen, and (iii) Tukey-Hanning. For the distance measurement we first conduct the adjacency matrix $A = (a_{ij})$, $a_{ij} = \max\{\mathbb{I}(w_{ij} > 0), \mathbb{I}(w_{ji} > 0)\}$. Then, we evaluate the geodesic distance d_{ij}^* . To gauge the distance between two pairs, we consider the three types of measures: (i) L^1 distance: $d_{ij,kl}^{*,1} = d_{ik}^* + d_{jl}^*$, (ii) L^2 (Euclidean) distance: $d_{ij,kl}^{*,2} = \sqrt{(d_{ik}^*)^2 + (d_{jl}^*)^2}$, and (iii) L^∞ distance: $d_{ij,kl}^{*,\infty} = \max\{d_{ik}^*, d_{jl}^*\}$. We set a bandwidth d_N to be the 25th-percentile of $\{d_{ij,kl}^*\}$. For the fixed-effect parameters, we report $\hat{\alpha}_{49}$ and $\hat{\eta}_{49}$ as representatives. We consider 1,000 sample repetitions. Table 2 summarizes the results.

Interpretations. Across designs, the PPMLE performs well in finite samples. We see small upward biases in $\hat{\lambda}_d$ and the fixed effects $(\hat{\alpha}_{49}, \hat{\eta}_{49})$, and mild downward biases in $\hat{\lambda}_o$, $\hat{\lambda}_w$, $\hat{\beta}_1$, and $\hat{\beta}_2$. For instance, the empirical biases are +0.0123 for $\hat{\lambda}_d$ and -0.0163 for $\hat{\lambda}_w$. These finite-sample biases attenuate as n increases; see Section 3.

For the main parameters $(\hat{\lambda}, \hat{\beta})$, our standard errors track the empirical standard deviations reasonably closely, albeit somewhat below them. As a result, coverage probabilities (CPs) are slightly below the 95% nominal rate. For example, under Parzen with an L^2 pair distance, the empirical STD and reported s.e. are (0.0266, 0.0229) for $\hat{\lambda}_d$ and (0.0137, 0.0120) for $\hat{\beta}_1$, with CPs of 0.933 and 0.897, respectively. Among kernel-distance combinations, we recommend the Parzen kernel with the L^2 pair distance for the main parameters, which delivers the most stable CPs across coefficients.

For the fixed-effect estimates $(\hat{\alpha}_{49}, \hat{\eta}_{49})$, the reported s.e. tend to be understated relative to the empirical STDs (e.g., $\hat{\alpha}_{49}$: STD 0.0845 vs s.e. 0.0299 under Parzen- L^∞), yielding CPs around 0.86–0.90. The gap narrows with larger n (Section 3). Within our design, Parzen with an L^∞ pair distance performs best for fixed effects. Intuitively, the L^∞ metric better captures the effective dependence radius in the FE direction by guarding against long-range pairwise links, which helps reduce s.e. underestimation.

As a rule of thumb, we recommend Parzen- L^2 for $(\hat{\lambda}, \hat{\beta})$ and Parzen- L^∞ for $(\hat{\alpha}, \hat{\eta})$. Monte Carlo uncertainty for a 95% CP with 1,000 replications is about 0.7 percentage points, so differences below ≈ 1 –2 points are not statistically meaningful, whereas the 4–8 point gaps we observe are. In applications, a slightly larger bandwidth (e.g., a +5–10 percentile increase) can further mitigate undercoverage for the fixed effects.

Table 2: Simulation results

	λ_d	λ_o	λ_w	β_1	β_2	α_{49}	η_{49}
Empirical bias	0.0123	-0.0004	-0.0163	-0.0011	-0.0001	0.0131	0.0044
Empirical STD	0.0266	0.0260	0.0355	0.0137	0.0130	0.0845	0.0850
s.e. (Bartlett, L^1)	0.0228	0.0226	0.0326	0.0117	0.0115	0.0283	0.0196
CP (Bartlett, L^1)	0.9180	0.9250	0.9050	0.9030	0.9110	0.8430	0.8290
s.e. (Bartlett, L^2)	0.0227	0.0227	0.0326	0.0117	0.0115	0.0241	0.0215
CP (Bartlett, L^2)	0.9190	0.9270	0.9120	0.9000	0.9160	0.8560	0.8630
s.e. (Bartlett, L^∞)	0.0227	0.0228	0.0326	0.0117	0.0116	0.0294	0.0224
CP (Bartlett, L^∞)	0.9180	0.9280	0.9120	0.8960	0.9070	0.8580	0.8830
s.e. (Parzen, L^1)	0.0228	0.0232	0.0330	0.0120	0.0118	0.0292	0.0216
CP (Parzen, L^1)	0.9310	0.9260	0.9200	0.9000	0.9180	0.8580	0.8710
s.e. (Parzen, L^2)	0.0229	0.0232	0.0331	0.0120	0.0118	0.0252	0.0228
CP (Parzen, L^2)	0.9330	0.9340	0.9280	0.8970	0.9180	0.8630	0.8910
s.e. (Parzen, L^∞)	0.0229	0.0233	0.0332	0.0120	0.0118	0.0299	0.0232
CP (Parzen, L^∞)	0.9320	0.9310	0.9290	0.8960	0.9160	0.8660	0.9030
s.e. (Tukey–Hanning, L^1)	0.0227	0.0228	0.0326	0.0118	0.0115	0.0284	0.0197
CP (Tukey–Hanning, L^1)	0.9170	0.9270	0.9080	0.9000	0.9140	0.8410	0.8220
s.e. (Tukey–Hanning, L^2)	0.0227	0.0229	0.0327	0.0118	0.0116	0.0242	0.0217
CP (Tukey–Hanning, L^2)	0.9140	0.9260	0.9080	0.9000	0.9170	0.8560	0.8680
s.e. (Tukey–Hanning, L^∞)	0.0227	0.0229	0.0327	0.0118	0.0116	0.0295	0.0224
CP (Tukey–Hanning, L^∞)	0.9220	0.9250	0.9130	0.8970	0.9150	0.8580	0.8840

4 Empirical Application

4.1 Basic setting

In this application, we quantify how network effects in bilateral trade flows shape effective trade costs through connections to third countries. Countries may reduce trade costs by routing or integrating shipments via third-party hubs, but congestion or capacity constraints at those hubs can also raise effective costs and dampen trade among neighboring countries.

We collect the international trade flows y_{ij} from the Center for International Data at UC Davis (<https://cid.ucdavis.edu/worldtradeflows>).¹¹ Then, y_{ij} denotes gross bilateral merchandise trade flows, including only *cross-border* transactions and excluding services, domestic intermediate transactions, and non-traded sectors. In the analysis, we therefore exclude the domestic trade (intra-trade flows y_{ii}) that are not directly observed in the data.¹²

¹¹Relevant explanations can be found in Feenstra et al. (2005).

¹²Although domestic trade is a well-defined theoretical concept, it is not directly observable in the data. When we define $y_{ii} = G_i - \sum_{j=1, j \neq i}^n y_{ij}$ using the country i 's GDP G_i , y_{ii} overwhelms the share since the GDP includes non-traded goods, services, and domestic intermediate transactions. Because our objective is to study how bilateral trade-cost shocks reshape the distribution of international trade relationships, we do not explicitly consider the intra flows y_{ii} . This approach is standard in the empirical trade and trade-network literature (Anderson and van Wincoop, 2003; Head and Mayer, 2014). While recent structural gravity literature emphasizes the importance of intra-national trade flows for identifying the border effect

For the covariates, we use the same set as in Helpman et al. (2008). FTA data are collected from WTO (https://www.wto.org/english/res_e/statis_e/statis_e.htm).

Understanding the evolution of global trade requires recognizing that international linkages are not static but are reshaped by major economic shifts. To illustrate how the structure and sources of network effects evolve, our main analysis focuses on two benchmark years that bracket the maturation of the global trade network: Period 1 (1997, the *developing network period* associated with active regional integration such as NAFTA) and Period 2 (2016, the *mature network period* marked by the expansion of global supply chains). Specifically, for $p \in \{1, 2\}$, let $y_{ij}^{(p)} \equiv y_{ij,t_p}$ denote the trade flow from j to i in year t_p , where $(t_1, t_2) = (1997, 2016)$. The number of countries included in each period is 142 and 147, respectively.¹³

Descriptive statistics. Table 3 reports descriptive statistics for the main variables in the two benchmark years used in our baseline analysis: Period 1 (1997) and Period 2 (2016). The mean bilateral trade flow (y_{ij}), measured in 2015 constant U.S. dollars, increases from about USD 0.38 million in Period 1 to about USD 0.72 million in Period 2, reflecting a robust expansion of global trade. At the extensive margin, the share of zero trade flows remains sizable but declines modestly (from about 30% to 28%), indicating that a substantial fraction of potential trade links remains inactive even as the network matures. Finally, the conditional mean among strictly positive flows (y_{ij}^+) rises from about USD 0.54 million to USD 1.00 million. These patterns also highlight the strong right-skewness of bilateral trade flows, with medians falling far below their respective means.

Countries' connectivity matrix To construct the countries' connectivity matrix W for each benchmark year, we use trade flows observed strictly prior to the outcome year to mitigate simultaneity. For Period $p \in \{1, 2\}$, let $\mathcal{T}_p$ denote the pre-period window used to build the network.¹⁴ For $i \neq j$, define the symmetric base weight

$$\tilde{w}_{ij}^{(p)} = \frac{1}{\#(\mathcal{T}_p)} \sum_{t \in \mathcal{T}_p} (y_{ij,t} + y_{ji,t}), \quad \tilde{w}_{ii}^{(p)} = 0,$$

(Yotov, 2022), our study focuses specifically on the reallocation effects within the international trade network.

¹³To provide a broader historical context and to assess robustness, Online Appendix 4.3 extends the analysis to two additional benchmark years: 1986 (early trade liberalization) and 2007 (the emergence of the China trade shock). The full list of country names is provided in Table 13.

¹⁴For the main periods, $\mathcal{T}_1 = \{1993, 1994, 1995, 1996\}$ and $\mathcal{T}_2 = \{2010, 2011, \dots, 2015\}$.

Table 3: Descriptive statistics (Main Periods)

Period	1 (1997)			2 (2016)		
	Mean	Median	STD	Mean	Median	STD
y_{ij} (Zero freq.)	380,573 (0.3007)	382	4,151,519	718,068 (0.2835)	865	7,174,976
y_{ij}^+	544,197	4,726	4,955,465	1,002,158	8,909	8,459,553
Distance	0.2864	0.2598	0.1801	0.2829	0.2543	0.1787
Border	0.0188	0.0000	0.1357	0.0186	0.0000	0.1352
Legal	0.3629	0.0000	0.4808	0.3673	0.0000	0.4821
Language	0.3100	0.0000	0.4625	0.3035	0.0000	0.4598
Colony	0.0110	0.0000	0.1042	0.0104	0.0000	0.1016
Currency	0.0100	0.0000	0.0994	0.0093	0.0000	0.0961
Islands	0.3662	0.0000	0.5450	0.3673	0.0000	0.5457
Landlock	0.3099	0.0000	0.5099	0.2993	0.0000	0.5028
FTA	0.0010	0.0000	0.0316	0.0107	0.0000	0.1030

Note: Period 1 corresponds to 1997 and Period 2 to 2016. Variable definitions follow Helpman et al. (2008).

and row-normalize to obtain

$$w_{ij}^{(p)} = \frac{\tilde{w}_{ij}^{(p)}}{\sum_{k \neq i} \tilde{w}_{ik}^{(p)}}, \quad w_{ii}^{(p)} = 0.$$

This yields a row-stochastic $W^{(p)}$ with a symmetric base $\tilde{W}^{(p)}$, consistent with Assumption 2.2. Each $w_{ij}^{(p)}$ admits a probability interpretation as the period-specific intensity of choosing j as i 's trade partner.

Table 4 summarizes how the connectivity network differs across Period 1 and Period 2. The network becomes denser (density rises from 0.76 to 0.91) and more diversified (both n^{HHI} and n^{E} increase). At the same time, the dispersion in degree declines, while the dispersion in high-intensity degree remains large, consistent with the presence of a small set of hub countries. Additional benchmark years and country-level statistics are reported in Online Appendix 4.

4.2 Estimation results

Table 5 summarizes how network effects differ across our two main benchmark years. We focus on the three network parameters $(\lambda_d, \lambda_o, \lambda_w)$, which govern how bilateral trade flows respond to changes in neighboring flows along the three network dimensions $I_n \otimes W$, $W \otimes I_n$,

Table 4: Summary network statistics (Main Periods)

	linear-in-means	bipartite	Period 1 (1997)	Period 2 (2016)
degree (deg)	149.0000	75.0000	118.9859	138.4354
Std(deg)	0.0000	0.0000	21.9047	12.1543
high-intensity degree (deg ⁺)	0.0000	0.0000	7.0493	7.3469
Std(deg ⁺)	0.0000	0.0000	19.1505	18.0075
HHI	0.0067	0.0133	0.1419	0.1182
Std(HHI)	0.0000	0.0000	0.1040	0.0881
n^{HHI}	149.0000	75.0000	9.7423	11.5994
PDE	1.0000	0.8628	0.5623	0.5921
Std(PDE)	0.0000	0.0000	0.0968	0.0924
n^{E}	149.0000	75.0000	17.8278	20.9272
$\varphi_{(2)}$	-0.0067	$\simeq 0$	0.5653	0.5553
$\varphi_{\min}$	-0.0067	-1	-0.5151	-0.4419
Density	1	0.5034	0.7560	0.9105

Note: $W^{(p)}$ is constructed from pre-period trade flows and row-normalized; see the text for details. Definitions of network statistics follow Online Appendix 4.

and $W \otimes W$.

Overall and key findings. As a key quantitative finding, allowing for these network spillovers substantially improves model fit relative to the conventional gravity equation with a purely iceberg-cost specification. The McFadden pseudo- R^2 reported in Table 5 equals 0.2848 in Period 1 and 0.1739 in Period 2. Since the McFadden pseudo- R^2 is defined relative to the conventional gravity benchmark, these values indicate sizable likelihood gains from incorporating third-party proximities and network interactions in both periods.

Implication. Because a well-measured cost specification is central to gravity models, the improvement in fit provides evidence that third-party proximities generate economically meaningful pair-specific heterogeneity in trade flows beyond what is absorbed by multilateral resistance terms (fixed effects). In this sense, the network channels in our model capture systematic patterns that would otherwise remain in the residual component of a conventional gravity regression. Consistent with this interpretation, the counterfactual analysis in Section 4.3 shows that these channels have distinct implications for trade reallocation that are absent under the conventional specification.

Period 1 (1997, developing network period). In Period 1, both λ_d and λ_o are positive and strongly significant, reflecting *complementarity* among trade flows along the exporter

and importer dimensions. Within our framework, $\lambda_d > 0$ implies that when an exporter deepens trade with some destinations, the weighted average of its flows to other destinations also tends to rise; analogously, $\lambda_o > 0$ captures complementarity among alternative suppliers serving a given destination. This pattern is consistent with reductions in trade frictions and institutional integration in the late 1990s (e.g., NAFTA and related regional arrangements) that facilitated multi-market participation and sourcing diversification. Bilateral flows thus become more mutually reinforcing within the trade network, and the indirect effects along $I_n \otimes W$ and $W \otimes I_n$ operate more like increasing returns to connectivity than crowding-out.

At the same time, λ_w is positive and precisely estimated, indicating economically important *third-party spillover effects*. A positive λ_w implies that the trade flow from j to i increases when countries that are simultaneously well connected to both i and j expand their trade, so that bilateral trade is amplified when partners share access to common hubs.

Period 2 (2016, mature network period). By Period 2, λ_d becomes smaller in magnitude and only weakly significant, while λ_o remains positive and highly significant but with a reduced magnitude relative to Period 1. This pattern suggests a partial stabilization of trade interdependencies within an increasingly dense and geographically diversified trade network. The modest value of λ_d points to a weaker role for substitution or complementarity along the “common-exporter” dimension, whereas the still positive λ_o implies that complementarity continues to operate primarily through the “common-importer” dimension.

The third-country parameter λ_w remains positive, sizeable, and precisely estimated in Period 2, indicating persistent hub-mediated spillovers in the mature trade network. Overall, the two-period comparison highlights strong and durable third-party effects once global supply chains are fully established, while the balance between complementarity across exporters and importers becomes more muted over time.

4.3 Counterfactual analysis

Motivated by the significant estimated spillover parameters, we investigate how changes in the economic environment alter the distribution of trade flows in our model and in the conventional gravity model. Accordingly, our framework provides a tool for ex-ante stress testing of trade networks. Because counterfactual trade shares are generated from the same estimated structure, policymakers can use the model to simulate hypothetical shocks—such as tariff escalations, decoupling strategies, or supply-chain disruptions—and trace how these shocks propagate through the network, reshaping import concentration and diversification.

Focusing on Phase 4, we consider a threefold increase in $\tau_{US,CN}^+$ and $\tau_{CN,US}^+$, thereby illus-

Table 5: Estimation Results (Main Periods)

Period	1 (1997)	2 (2016)
λ_d	0.3002 (0.0362)	0.1370 (0.0712)
λ_o	0.3510 (0.0373)	0.2830 (0.0649)
λ_w	0.3354 (0.0336)	0.5734 (0.0418)
Distance	-1.5184 (0.2449)	-1.7511 (0.2849)
Border	0.9973 (0.2134)	0.9402 (0.1575)
Legal	0.2029 (0.0682)	0.1337 (0.0698)
Language	-0.0159 (0.0547)	0.0225 (0.0697)
Colony	0.0000 (0.0985)	0.1386 (0.0871)
Currency	-0.0682 (0.4260)	-0.0622 (0.3185)
Islands	-8.0810 (0.0784)	-6.3638 (0.0639)
Landlock	-7.2785 (0.0649)	-10.5669 (0.0482)
FTA	0.4123 (0.1717)	0.4968 (0.0851)
# of observations	20,022	21,462
Log-likelihood	-1,323,341,030	-3,009,628,447
McFadden's R^2	0.2848	0.1739

Note: Standard errors are computed using a Parzen kernel with an L^2 -based pair distance and are reported in parentheses. Period 1 corresponds to 1997 and Period 2 to 2016. The McFadden pseudo- R^2 is defined as $1 - \hat{\ell}_N / \hat{\ell}_N^{\text{trad}}$, where $\hat{\ell}_N^{\text{trad}}$ denotes the log-likelihood of the conventional gravity benchmark. Variable definitions and additional implementation details are reported in the Online Appendix.

trating the recent US-China trade war as a counterfactual scenario. Specifically, we compare predicted trade flows using our estimates ($\hat{\cdot}$) with those obtained from the counterfactual scenario ($\tilde{\cdot}$). We set the elasticity of substitution to $\varrho = 5$, following Anderson and van Wincoop (2003).¹⁵ Based on the counterfactual fixed-effect components $\tilde{\alpha}$ and $\tilde{\eta}$ obtained by (2.8), we generate the counterfactual trade flows $\tilde{\mu}$.¹⁶

Next, we compute the estimated and counterfactual shares to understand changes in trade flows from different economic environments. In both models, importer budget G_i is taken as given/absorbed by fixed effects, so the counterfactual is not designed to predict aggregate trade volumes. The economically meaningful adjustment margin is the reallocation of a fixed import budget across partners, which is naturally summarized by import shares. Hence, the counterfactual is best interpreted as a redistribution exercise: the shock primarily changes the composition of each importer's sourcing across partners, rather than providing a forecast of aggregate trade volume. For this reason, we summarize equilibrium adjustments using changes in import shares, which differ across importers and isolate reallocation across partners.¹⁷ For each $i = 1, \dots, n$,

$$\hat{s}_{ij} = \frac{\hat{\mu}_{ij}}{\sum_{k=1, k \neq i} \hat{\mu}_{ik}} \text{ and } \tilde{s}_{ij} = \frac{\tilde{\mu}_{ij}}{\sum_{k=1, k \neq i} \tilde{\mu}_{ik}} \text{ for } j \neq i.$$

For comparison, we also compute the estimated and counterfactual shares $\hat{s}_{ij}^{\text{con}}$ and $\tilde{s}_{ij}^{\text{con}}$ from the conventional gravity model. We report the results by selecting 10 countries with four categories: (i) Category 1 (Hub countries having high degree with low concentration): U.S., Germany, and Japan, (ii) Category 2 (Countries with structural changes): China and South Korea, (iii) Category 3 (Countries with high concentration): Canada and Mexico, and (iv) Category 4 (Countries with low concentration): Sudan, Egypt, and Kenya.

¹⁵Head and Mayer (2014) also set the value of ϱ around 5 (i.e., $\varrho = 5.03$) as their preferred estimate from their meta analysis.

¹⁶In detail, we generate $\tilde{\mu}$ by the following steps.

- **Step 1.** Evaluate $\hat{\tau}_{ij}^+ = \exp\left(\frac{1}{1-\varrho} \cdot x'_{ij} \hat{\beta}\right)$ using $\hat{\beta}$ and $\varrho = 5$.
- **Step 2.** Construct $\tilde{\tau}_{ij}^+ = \hat{\tau}_{ij}^+ + \Delta_{ij}^+$, where $\Delta_{ij}^+ = \begin{cases} 2\hat{\tau}_{ij}^+ & \text{if } ij \in \{(\text{US}, \text{CN}), (\text{CN}, \text{US})\}, \\ 0 & \text{otherwise.} \end{cases}$
- **Step 3.** Generate $\tilde{\mu}$ with $\tilde{\alpha}$ and $\tilde{\eta}$ as the data generating process in Sec 3.3.

¹⁷Both our model and the conventional gravity framework (Anderson and van Wincoop, 2003) embed a CES demand system with multilateral resistance. Our counterfactual perturbs only the targeted bilateral trade-cost components and recomputes equilibrium objects within the same estimated structure, so the exercise is naturally interpreted as tracing *reallocation* across partners. Reporting changes in levels μ_{ij} may conflate two margins: changes in the overall scale of importer i 's imports and changes in the *composition* of imports across partners. Import shares provide a scale-free, directly comparable summary of the equilibrium adjustment across models and countries.

Model’s mechanism. By increasing $\tau_{\text{US,CN}}^+$ and $\tau_{\text{CN,US}}^+$, we study how a bilateral cost shock propagates through the equilibrium mapping and ultimately reshapes the import shares $\{s_{ij}\}_{j \neq i}$.

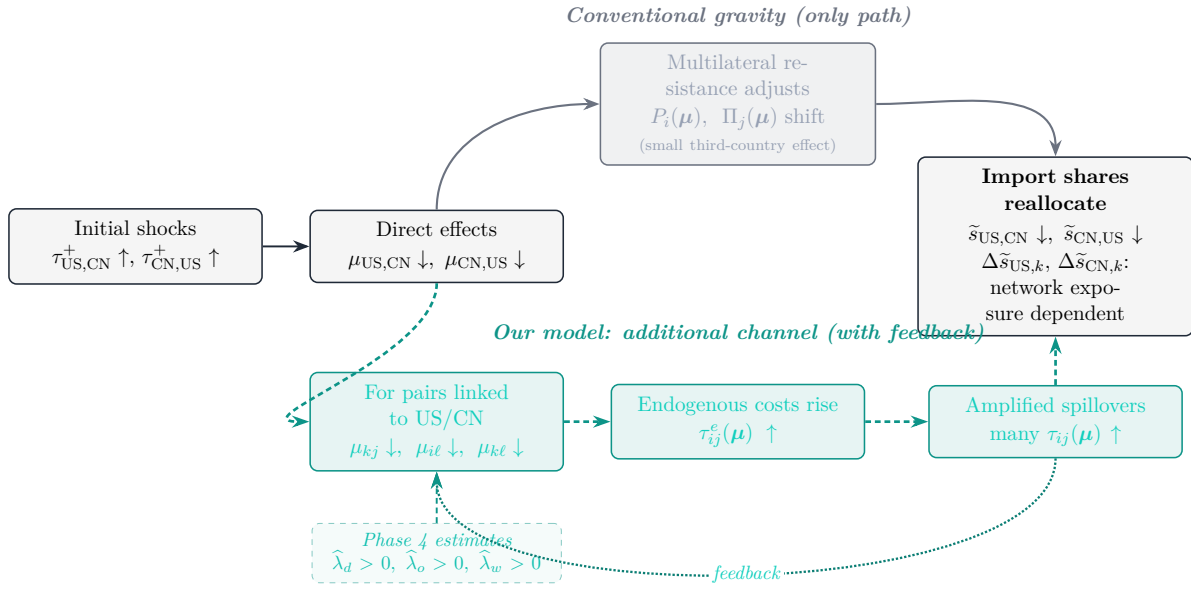
- Conventional gravity model ($\lambda = 0$):
 - Direct effect: These changes directly lead to decreasing $\mu_{\text{US,CN}}$ and $\mu_{\text{CN,US}}$.
 - Via the multilateral resistance terms: These shocks change the multilateral resistance terms $\{P_i, \Pi_j\}_{i,j=1}^n$ through price-index/expenditure-share re-optimization, implying adjustments in third-country flows μ_{kl} (for $k \neq i$ or $l \neq j$) through the multilateral resistance terms (the magnitude depends on substitution patterns and baseline trade shares).¹⁸
- Our model’s additional channels (if $\hat{\lambda}_d > 0$, $\hat{\lambda}_o > 0$, and $\hat{\lambda}_w > 0$):
 - Endogenous network trade-cost channel: The increase in $\tau_{\text{US,CN}}^+$ and $\tau_{\text{CN,US}}^+$ reduces the targeted flows, reallocates $\boldsymbol{\mu}$ in equilibrium, and thereby changes $\tau_{kl}^e(\boldsymbol{\mu})$ for many non-targeted pairs kl . Hence, trade flows μ_{kl} can change even when τ_{kl}^+ is held fixed.
 - Multilateral resistance amplification: Because network costs $\tau_{ij}^e(\boldsymbol{\mu})$ respond endogenously to the reallocation of $\boldsymbol{\mu}$, the adjustment in $\{P_i(\boldsymbol{\mu}), \Pi_j(\boldsymbol{\mu})\}$ is typically larger and more system-wide than under $\lambda = 0$, generating additional propagation of the bilateral shock to third-country trade flows. This is because $\tau^e(\boldsymbol{\mu})$ shifts the entire vector of bilateral trade costs that enters the CES price indices.

Figure 3 illustrates the model’s main mechanism as a diagram.

Summary of the key findings. While higher U.S.–China trade costs naturally reshape the import composition of the two countries involved, our network model implies substantial adjustments in the import shares of third countries. In contrast, under the conventional gravity framework, these third-country effects are largely absent. This difference arises because, in our model, trade flows are jointly determined through the network multiplier matrix $\mathbf{S}^{-1}$, so that a bilateral shock propagates through origin-, destination-, and hub-based linkages.

¹⁸The signs of ΔP_i , $\Delta \Pi_j$, and $\Delta \mu_{kl}$ are generally ambiguous and depend on substitution patterns and initial trade shares.

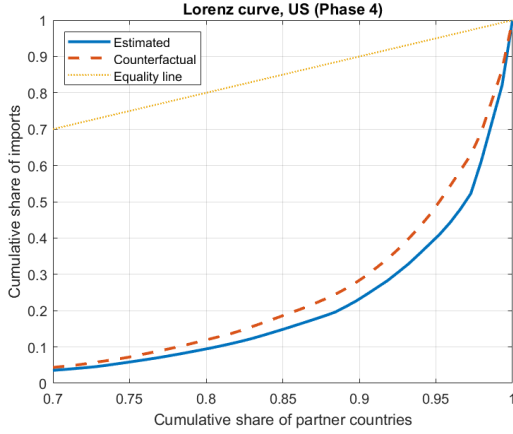
Figure 3: Model's mechanism



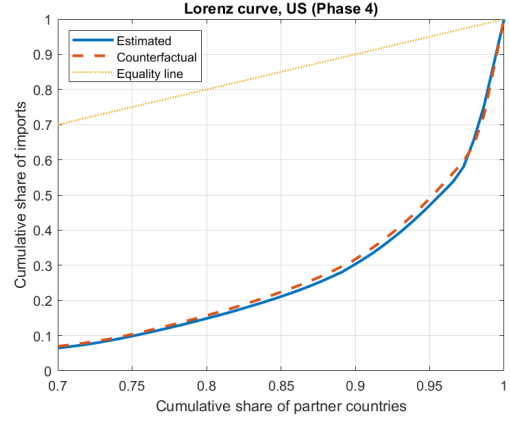
Note: Solid gray arrows indicate channels present in the conventional gravity model (i.e., $\lambda = 0$). Dashed teal arrows represent additional propagation mechanisms implied by our estimated network spillovers ($\hat{\lambda}_d > 0, \hat{\lambda}_o > 0, \hat{\lambda}_w > 0$). The initial bilateral shock raises exogenous trade-cost components for the targeted pair, reducing the corresponding flows. In our model, this reallocation alters endogenous network trade costs $\tau_{ij}^e(\mu)$, amplifies multilateral resistance adjustments, and transmits the shock to non-targeted pairs through a feedback loop. The “Import shares reallocate” box reports the counterfactual object: changes in import shares $\tilde{s}_{ij}$ for $j \neq i$.

Figure 4: Lorenz curve (Category 1: Hub economies)

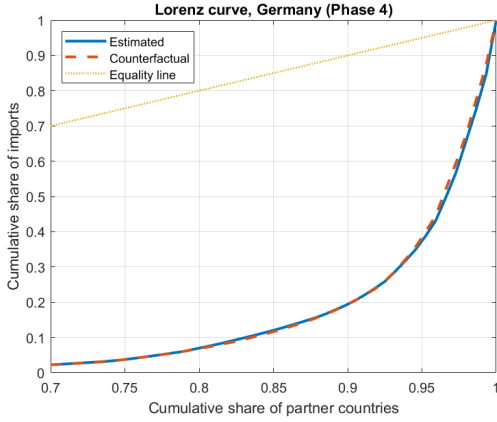
(a) U.S. (Our model)



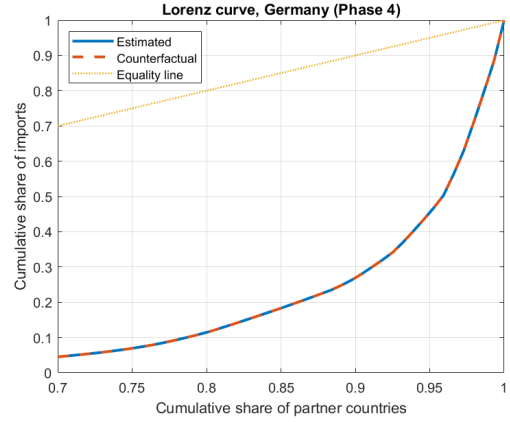
(b) U.S. (Conventional model)



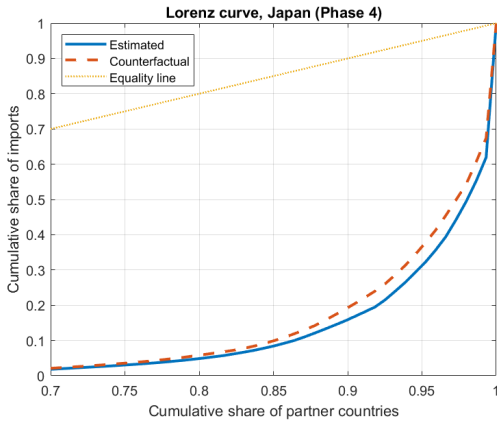
(c) Germany (our model)



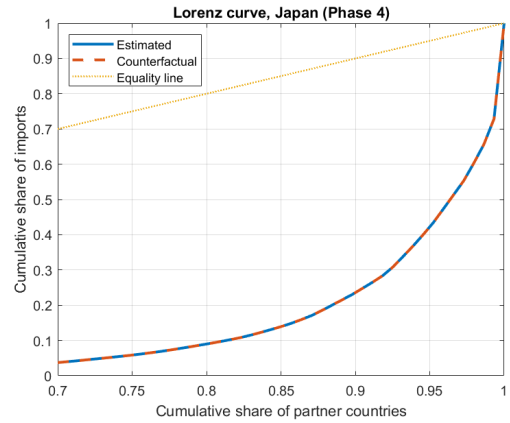
(d) Germany (Conventional model)



(e) Japan (our model)



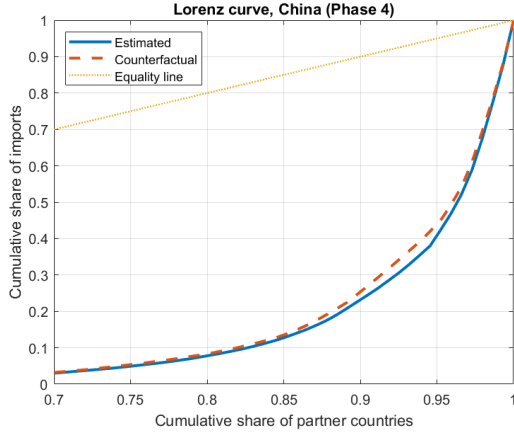
(f) Japan (Conventional model)



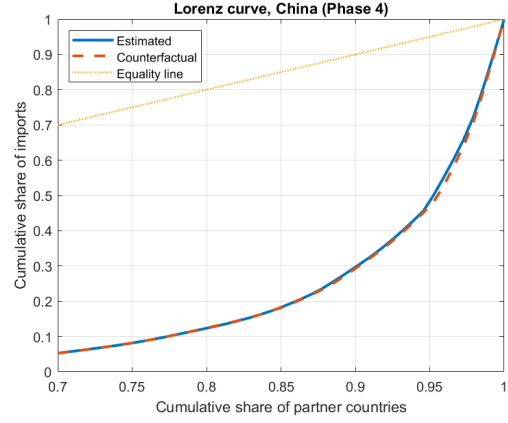
Note: The Gini coefficients from the Lorenz curves are as follows: (i) For Panels (a) and (b), $\text{Gini}(\hat{s}_{\text{US},\cdot}) = 0.8574$, $\text{Gini}(\tilde{s}_{\text{US},\cdot}) = 0.8304$, $\text{Gini}(\hat{s}_{\text{US},\cdot}^{\text{con}}) = 0.8146$, $\text{Gini}(\tilde{s}_{\text{US},\cdot}^{\text{con}}) = 0.8074$; (ii) For Panels (c) and (d), $\text{Gini}(\hat{s}_{\text{DE},\cdot}) = 0.8726$, $\text{Gini}(\tilde{s}_{\text{DE},\cdot}) = 0.8712$, $\text{Gini}(\hat{s}_{\text{DE},\cdot}^{\text{con}}) = 0.8331$, $\text{Gini}(\tilde{s}_{\text{DE},\cdot}^{\text{con}}) = 0.8331$; (iii) For Panels (e) and (f), $\text{Gini}(\hat{s}_{\text{JP},\cdot}) = 0.8996$, $\text{Gini}(\tilde{s}_{\text{JP},\cdot}) = 0.8849$, $\text{Gini}(\hat{s}_{\text{JP},\cdot}^{\text{con}}) = 0.8571$, $\text{Gini}(\tilde{s}_{\text{JP},\cdot}^{\text{con}}) = 0.8571$.

Figure 5: Lorenz curve (Category 2: Structural-changed economies)

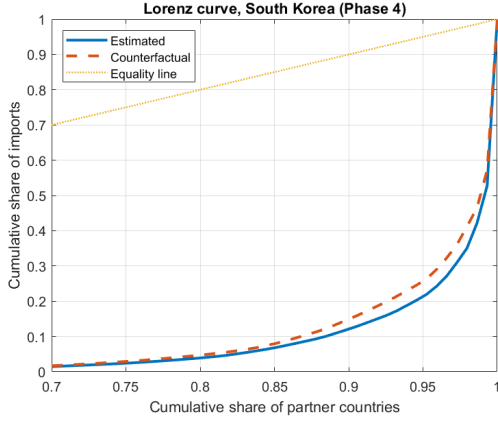
(a) China (Our model)



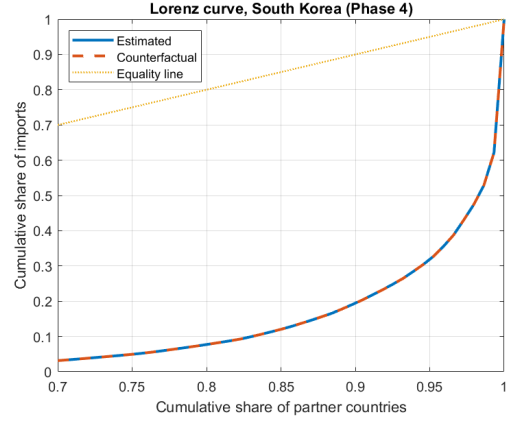
(b) China (Conventional model)



(c) South Korea (our model)



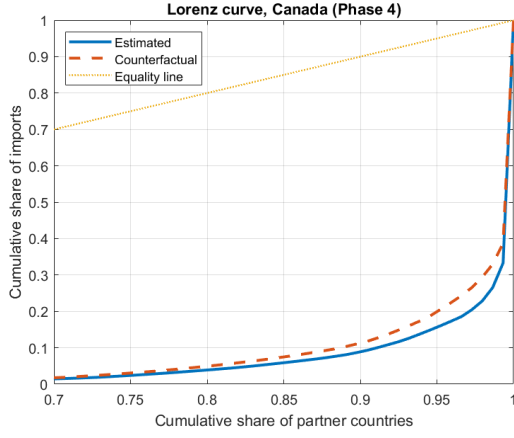
(d) South Korea (Conventional model)



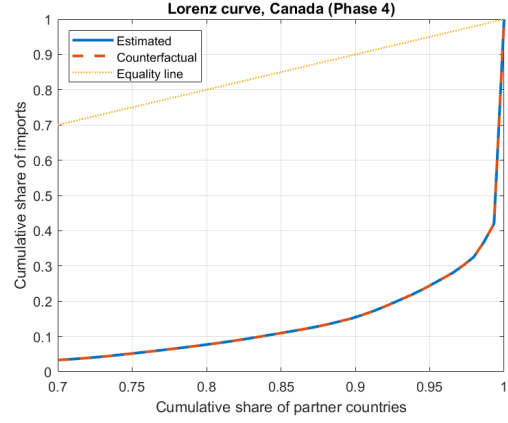
Note: The Gini coefficients from the Lorenz curves are as follows: (i) For Panels (a) and (b), $\text{Gini}(\hat{s}_{\text{CN},.}) = 0.8601$, $\text{Gini}(\tilde{s}_{\text{CN},.}) = 0.8516$, $\text{Gini}(\hat{s}_{\text{CN},.}^{\text{con}}) = 0.8232$, $\text{Gini}(\tilde{s}_{\text{CN},.}^{\text{con}}) = 0.8250$; (ii) For Panels (c) and (d), $\text{Gini}(\hat{s}_{\text{KR},.}) = 0.9229$, $\text{Gini}(\tilde{s}_{\text{KR},.}) = 0.9102$, $\text{Gini}(\hat{s}_{\text{KR},.}^{\text{con}}) = 0.8827$, $\text{Gini}(\tilde{s}_{\text{KR},.}^{\text{con}}) = 0.8827$.

Figure 6: Lorenz curve (Category 3: Highly concentrated economies)

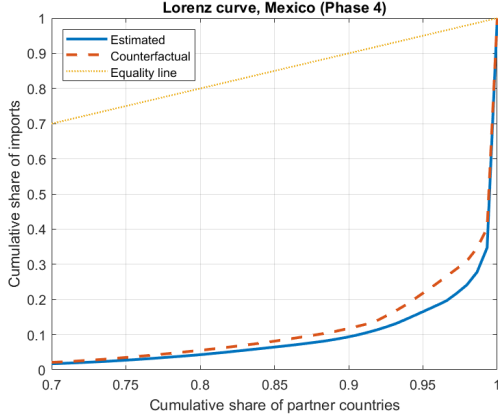
(a) Canada (Our model)



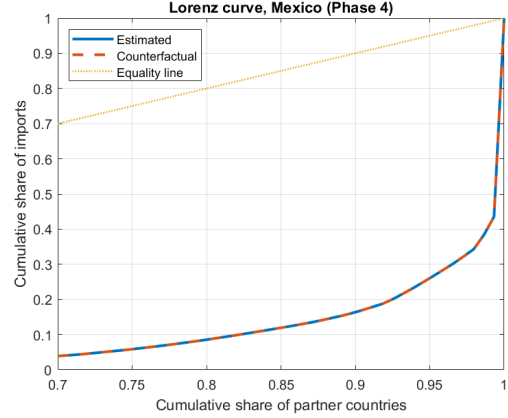
(b) Canada (Conventional model)



(c) Mexico (our model)



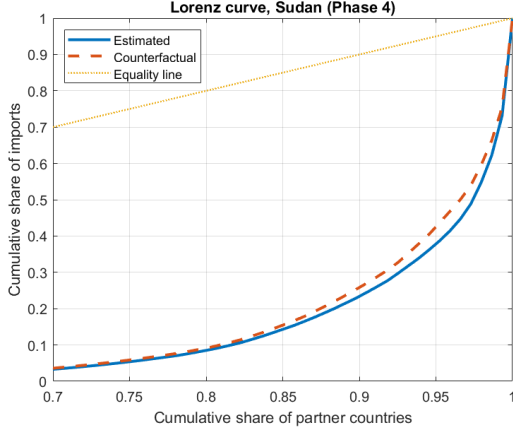
(d) Mexico (Conventional model)



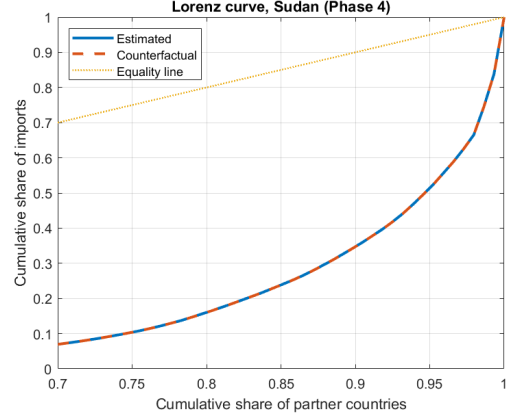
Note: The Gini coefficients from the Lorenz curves are as follows: (i) For Panels (a) and (b), $\text{Gini}(\hat{s}_{CA,.}) = 0.9399$, $\text{Gini}(\tilde{s}_{CA,.}) = 0.9260$, $\text{Gini}(\hat{s}_{CA,.}^{\text{con}}) = 0.9019$, $\text{Gini}(\tilde{s}_{CA,.}^{\text{con}}) = 0.9019$; (ii) For Panels (c) and (d), $\text{Gini}(\hat{s}_{MX,.}) = 0.9359$, $\text{Gini}(\tilde{s}_{MX,.}) = 0.9206$, $\text{Gini}(\hat{s}_{MX,.}^{\text{con}}) = 0.8945$, $\text{Gini}(\tilde{s}_{MX,.}^{\text{con}}) = 0.8945$.

Figure 7: Lorenz curve (Category 4: Low-concentration economies)

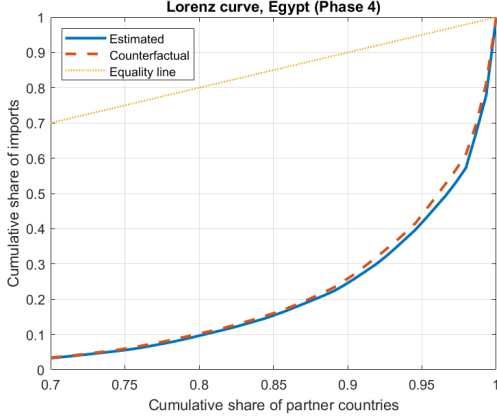
(a) Sudan (Our model)



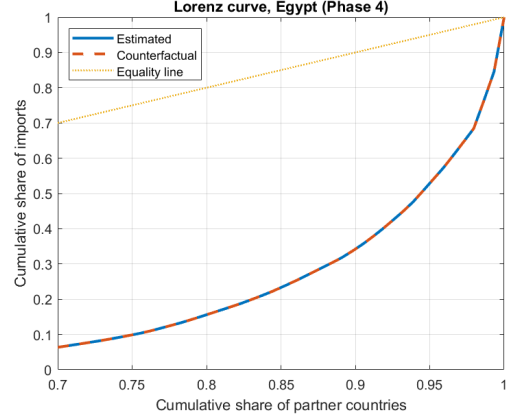
(b) Sudan (Conventional model)



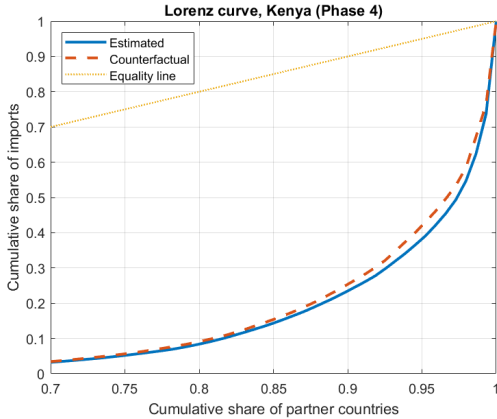
(c) Egypt (our model)



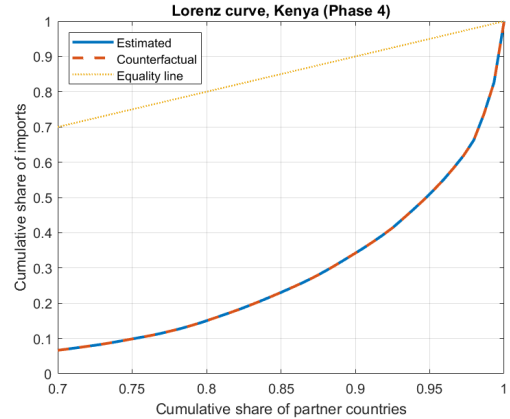
(d) Egypt (Conventional model)



(e) Kenya (our model)



(f) Kenya (Conventional model)



Note: The Gini coefficients from the Lorenz curves are as follows: (i) For Panels (a) and (b), $\text{Gini}(\hat{s}_{SS,\cdot}) = 0.8660$, $\text{Gini}(\tilde{s}_{SS,\cdot}) = 0.8544$, $\text{Gini}(\hat{s}_{SS,\cdot}^{\text{con}}) = 0.8010$, $\text{Gini}(\tilde{s}_{SS,\cdot}^{\text{con}}) = 0.8010$; (ii) For Panels (c) and (d), $\text{Gini}(\hat{s}_{EG,\cdot}) = 0.8568$, $\text{Gini}(\tilde{s}_{EG,\cdot}) = 0.8497$, $\text{Gini}(\hat{s}_{EG,\cdot}^{\text{con}}) = 0.8029$, $\text{Gini}(\tilde{s}_{EG,\cdot}^{\text{con}}) = 0.8029$; (iii) For Panels (e) and (f), $\text{Gini}(\hat{s}_{KE,\cdot}) = 0.8657$, $\text{Gini}(\tilde{s}_{KE,\cdot}) = 0.8559$, $\text{Gini}(\hat{s}_{KE,\cdot}^{\text{con}}) = 0.8054$, $\text{Gini}(\tilde{s}_{KE,\cdot}^{\text{con}}) = 0.8054$.

Category 1: Hub economies. For the U.S., the increase in U.S.–China trade costs results in an almost complete collapse in China’s import share (from $\hat{s}_{\text{US,CN}} = 0.1790$ to $\tilde{s}_{\text{US,CN}} = 0.0024$). In our model, this loss is absorbed by a broad set of alternative suppliers (e.g., Germany, Japan, the UK, France, Canada, etc.), resulting in pronounced declines in the Gini coefficient. By contrast, under conventional gravity, China’s share also falls (from $\hat{s}_{\text{US,CN}}^{\text{con}} = 0.0951$ to $\tilde{s}_{\text{US,CN}}^{\text{con}} = 0.0013$), but the adjustment is concentrated among a small number of large partners, leaving overall concentration nearly unchanged.

For Germany and Japan—countries not directly targeted by the bilateral shock—the conventional gravity counterfactual yields virtually no change in import shares. In sharp contrast, our model predicts sizable reallocations, reflecting the propagation of the US–China shock through the countries’ network connectivity. These effects on Japan are greater than those on Germany due to Japan’s connectivity to the U.S. and China.

Category 2: Structural-changed economies. China and Korea exhibit particularly strong responses. Both economies initially exhibit a high degree of concentration among a small set of partners. Following the shock, our model predicts substantial diversification, as indicated by declining Gini coefficients. These adjustments are largely absent under conventional gravity.

Category 3: Highly concentrated economies. Canada and Mexico initially rely heavily on the U.S. market. In the counterfactual, the reorganization of U.S. import demand partially unwinds this dependence, leading to lower concentration and greater diversification of import shares.

Category 4: Low-concentration economies. Even for relatively peripheral economies, whose exogenous bilateral trade costs are left unchanged, our model predicts noticeable redistribution of import shares.

4.4 Discussion

Our counterfactual analysis of the U.S.–China trade war highlights that trade policy shocks operate not only through direct bilateral channels but also through the countries’ connectivity network. In our framework, a change in bilateral trade costs alters equilibrium trade flows through origin-based, destination-based, and hub-mediated linkages, generating reallocations that extend well beyond the directly targeted country pair. This mechanism produces

rich distributional responses in import shares that are largely absent under a conventional gravity model.

First, trade policy evaluation must be network-aware. While the increase in U.S.–China trade costs naturally reshapes the import composition of the two countries involved, our results show that third countries experience substantial reallocations of import shares even when their (exogenous) bilateral trade costs remain unchanged. In particular, hub economies such as Japan—neither of which is directly targeted by the shock—exhibit sizable changes in import shares under our model (In contrast, conventional gravity predicts little response in import shares.). This contrast implies that ignoring network propagation risks severely understates the scope of policy effects.

Second, hub countries play a systemic role in transmitting trade shocks. Countries with strong network connectivity act as conduits through which bilateral shocks are redistributed across the global system. In the U.S., the decline in China’s import share is offset by a broad range of alternative suppliers, leading to a marked decrease in import concentration. This diversification arises precisely because trade flows are jointly determined through network interactions. From a policy perspective, this implies that shocks or interventions affecting highly connected hubs—whether through tariffs, sanctions, or industrial policy—can generate disproportionately large spillovers, both stabilizing and destabilizing, for the rest of the network.

Third, network effects shape the distributional consequences of trade shocks. Our comparison of Lorenz curves shows that the same policy shock can generate distinct concentration responses across countries, depending on their initial positions in their connectivity networks. Economies with initially concentrated import structures, such as Canada and Mexico, experience substantial diversification as dependence on the U.S. market is partially unwound. By contrast, peripheral economies with low initial concentration—such as Sudan, Egypt, and Kenya—also experience noticeable redistribution of import shares, despite facing no direct change in bilateral trade costs. Without an explicit network-based framework, trade policy can reallocate gains and losses among partners in highly difficult-to-anticipate ways.

5 Conclusion

The gravity equation has long served as a foundation for analyzing origin–destination flows, tracing back to Isard (1954) and Tinbergen (1962). Although these early formulations were groundbreaking in highlighting the role of geographic distance in trade, they remained within an exogenous framework. This bilateral focus isolated pairwise effects—such as shared bor-

ders, common languages, or trade agreements—but failed to capture how indirect and higher-order connections among countries jointly shaped trade outcomes. Recognizing the growing importance of trade networks, we developed a microfoundation-based specification and econometric framework that endogenized trade costs as a function of network structure, thereby making them both endogenous and interdependent, thereby amplifying heterogeneity across trade pairs.

From a theoretical standpoint, we derived the specification from microfoundations relying on a spatial autoregressive structure that naturally captured multilateral interdependence. Our framework departed from the traditional iceberg-cost assumption by allowing trade costs to reflect network linkages, in which countries leveraged their trade connections as a resource.

Methodologically, we extended the Poisson pseudo maximum likelihood estimator (PPMLE) with accommodating heteroskedasticity- and autocorrelation-robust standard errors to ensure valid inference in the presence of arbitrary correlation in the error structure. To implement PPMLE, we develop a spectral algorithm for computing transformations by the network multiplier matrix. Rather than forming and inverting a large-dimensional matrix for the network multiplier matrix, our algorithm utilizes the eigendecomposition of the row-normalized countries’ connectivity matrix to simultaneously diagonalize the three Kronecker-type network operators for cross-destination, cross-origin, and joint origin–destination linkages. As a result, linearly transforming a vector by the network multiplier matrix can be implemented via simple elementwise operations in the eigenbasis. This approach substantially decreases the computational burden compared to the naive inverse-based approach.

Empirically, we identified significant and distinct patterns of network effects across four key phases of global trade. In light of the recent U.S.–China trade war, our counterfactual analysis further demonstrates that a bilateral increase in trade costs induces substantial reallocation of import shares across a wide range of third countries when network interactions are accounted for. These results underscore the importance of viewing the global trading system as an interconnected network rather than a collection of independent bilateral relationships.

As final remarks, we note two possible extensions. First, our framework might be extended to interrelated sectors and countries. As an initial point of view, we focused on aggregate goods rather than sector-specific goods. As Caliendo and Parro (2015) point out, however, sectors are also interrelated. In this case, a possible data structure might be $y_{ip,jq}$ representing flows from country j ’s sector q to country i ’s sector p . Second, although our primary focus was on international trade flows, our proposed framework might be broadly applicable to other types of origin-destination flows (commuting or migration flows). In

particular, our model would be useful for explaining systemized flows when cross-sectional units are interconnected, and the formation of flows is heavily influenced by their interconnectedness.

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